

INTERNSHIP PROJECT REPORT

City Scale Retrofitting Analysis for Ahmedabad

Submitted by

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Internship Organization

Swa Consultancy



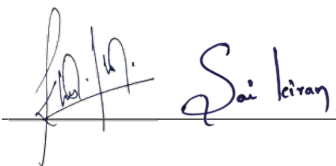
Academic Year: 2026

Project Repository (GitHub): City Scale Retrofitting Analysis for Ahmedabad

Certificate

This is to certify that **Kavya Durga Sri Gullapalli** and **Srushti Dandekar** have successfully completed the internship project titled **City Scale Retrofitting Analysis for Ahmedabad** under the guidance of **Mr Saikiran** and **Miss Khushi Thapa** during the internship period from **14th May** to **15th July 2026**.

The work presented in this report is original and has been completed as part of the internship requirements.

Guide Signature:  Sai Kiran

Organization Representative: Viraj Shah

Date: 20-07-2026

Declaration

We hereby declare that this project report entitled "**City Scale Retrofitting Analysis on Ahmedabad**" is my original work carried out during my internship. The information and data presented in this report are authentic to the best of my knowledge and belief.

Student Name: Kavya Durga Sri, Srushti Dandekar

Signature: G. Kavya Durga Sri 

Date: 20-07-2026

Acknowledgement

I express my sincere gratitude to my internship organization for providing me with the opportunity to work on this project. I thank my project mentor, faculty guide, and colleagues for their continuous guidance, encouragement, and support throughout the internship. Their valuable suggestions and feedback greatly contributed to the successful completion of this project.s my sincere gratitude to my internship organization for providing me with the opportunity to work on this project.

Author Contributions

This project was carried out collaboratively by **Kavya Durga Sri Gullapalli** and **Srushti Dandekar** under the guidance of the project mentor at SWA Consultancy. Both team members actively participated throughout the internship in literature review, technical discussions, project planning, documentation, and presentations. The major individual contributions are summarized below.

Kavya Durga Sri

- Participated in the initial selection and exploration of study areas within Ahmedabad for Urban Building Energy Modeling (UBEM).
- Processed the final GIS datasets for Navrangpura and Thaltej, including attribute cleaning, building height estimation, footprint area calculation, total floor area estimation, building classification, and preparation of CityBES-compatible GeoJSON files.
- Explored multiple weather data repositories and identified the Ahmedabad EPW weather file selected for EnergyPlus and CityBES simulations.
- Studied the CityBES documentation to understand the required `additional_json` structure and operational parameters.
- Developed Python scripts to automatically integrate Ahmedabad-specific `additional_json` parameters into every building feature of the GeoJSON datasets.
- Verified the final GeoJSON datasets for compatibility with CityBES and EnergyPlus.
- Contributed to the preparation of the project report, presentation slides, and technical documentation.

Srushti Dandekar

- Participated in the initial selection and exploration of study areas within Ahmedabad for Urban Building Energy Modeling (UBEM).
- Processed GIS datasets for multiple Ahmedabad regions using QGIS, including attribute preparation and GeoJSON development. However, due to an unexpected QGIS project crash, the processed files could not be recovered and were therefore not used in the final dataset preparation.
- Contributed to the literature review on Urban Building Energy Modeling (UBEM), CityBES, EnergyPlus, retrofit analysis, and sustainable building technologies.

- Explored multiple weather data sources and evaluated suitable EPW datasets for Ahmedabad.
- Participated in discussions related to CityBES workflow, simulation inputs, and retrofit methodologies.
- Contributed to the preparation of the project report, presentation slides, and technical documentation.

Joint Contribution

- Studied the CityBES framework, EnergyPlus simulation workflow, and Urban Building Energy Modeling concepts.
- Participated in regular discussions with project mentors regarding data preparation, simulation methodology, and project progress.
- Prepared weekly progress presentations and technical documentation throughout the internship.
- Collaboratively reviewed project outputs and finalized the internship report and presentation.

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Abstract

This internship project focuses on the development of an Urban Building Energy Modeling (UBEM) workflow for residential buildings in Ahmedabad using City Building Energy Saver (CityBES). The objective of the project is to prepare accurate city-scale building datasets that can be used for evaluating building energy consumption and assessing potential retrofit strategies for improving energy efficiency.

The project involved collecting building footprint data from OpenStreetMap, processing and cleaning the datasets using QGIS, estimating building heights and other geometric properties, and generating standardized GeoJSON files compatible with the CityBES platform. Additional building parameters such as HVAC system type, cooling setpoints, heating schedules, and climate-specific operational assumptions were incorporated using Python scripting based on Ahmedabad residential building characteristics.

Various GIS techniques including spatial data extraction, attribute processing, building classification, GeoJSON preparation, and Python-based dataset enhancement were employed throughout the project. Ahmedabad-specific operational parameters were incorporated into the datasets through automated GeoJSON modification using Python, and the final datasets were validated for compatibility with the CityBES platform to facilitate city-scale building energy simulations.

Baseline and retrofit simulations were performed using EnergyPlus for two residential districts, Navrangpura and Thaltej. The retrofit strategy included improvements to the building envelope through Polyurethane Foam (PUF) insulation, Vacuum Insulated Glazing (VIG), enhanced cooling system efficiency, and reduced lighting power density. The proposed retrofit measures achieved reductions of approximately **16.1%** and **17.4%** in annual site energy consumption for Navrangpura and Thaltej, respectively, while cooling energy demand decreased by approximately **30.5%** and **32.2%**. These results demonstrate the effectiveness of the proposed retrofit strategies in improving the energy performance of residential buildings under Ahmedabad's hot climatic conditions.

The developed workflow provides a systematic methodology for preparing standardized Urban Building Energy Modeling (UBEM) datasets for Indian cities and establishes a foundation for future analyses involving energy benchmarking, large-scale retrofit evaluation, and sustainable urban planning.



Figure 1: Before and After Retrofitting

1 Introduction

Buildings account for a significant portion of global energy consumption and greenhouse gas emissions. Rapid urbanization in Indian cities has increased the demand for reliable tools that can evaluate building energy performance at the city scale and support sustainable urban planning.

Urban Building Energy Modeling (UBEM) enables simulation of energy consumption for thousands of buildings simultaneously by integrating geographic information systems (GIS), building characteristics, climate data, and energy simulation engines. Such models assist planners and researchers in evaluating retrofit measures, estimating energy savings, and developing strategies for reducing urban energy demand.

The objective of this internship was to contribute to the preparation of Ahmedabad residential building datasets compatible with the City Building Energy Saver (CityBES) platform. The work primarily involved collecting and processing GIS datasets, extracting and validating building attributes, estimating building heights, classifying buildings based on footprint area, preparing standardized GeoJSON datasets and incorporating Ahmedabad-specific operational parameters using Python for building energy simulation.

The project demonstrates how GIS-based data processing can be integrated with building energy simulation tools to support city-scale sustainability studies and evidence-based decision making.

2 Organization Profile

Organization Name: Swa Consultancy Pvt. Ltd.

2.1 Company Overview

Swa Consultancy is an Ahmedabad-based engineering consultancy specializing in thermal, acoustic, and passive fire insulation design for industrial, commercial, and institutional projects. Headquartered at *303, Safal Prelude, Prahladnagar, Ahmedabad, Gujarat*, the company provides engineering solutions focused on improving energy efficiency, controlling noise, and ensuring compliance with national building and fire safety standards.

The company's name is derived from the Sanskrit word "*Swa*", meaning "*self*", reflecting its philosophy of taking complete ownership of engineering challenges from initial design calculations to final site verification.

2.2 Vision

To promote sustainable and energy-efficient engineering solutions through innovative insulation design and urban energy analysis.

2.3 Mission

- Deliver technically sound and cost-effective insulation solutions.
- Improve building energy efficiency and occupant comfort.
- Support sustainable infrastructure through advanced engineering practices.
- Ensure compliance with national building and fire safety standards.

2.4 Core Service Areas

SWA Consultancy primarily operates in the following engineering domains:

1. Thermal Insulation Design

Design of insulation systems for hot and cold pipes, HVAC ducts, storage tanks, and building envelopes to reduce heat loss, prevent condensation, and improve energy efficiency in accordance with the Energy Conservation Building Code (ECBC).

2. Acoustic Insulation Design

Design of soundproofing and noise-control systems for industrial facilities, commercial buildings, hotels, offices, HVAC ducts, and DG rooms using materials such as rockwool, glasswool, and acoustic boards.

3. Passive Fire Insulation Design

Design of passive fire protection systems for industrial plants and commercial buildings in compliance with the National Building Code (NBC) and relevant Indian Standards (IS).

4. Insulation Audits

Assessment of existing insulation systems to identify heat losses, noise leakage, and code-compliance issues, followed by retrofit recommendations and energy-saving estimates.

2.5 Profile and Experience

SWA Consultancy has over **35 years of combined experience** in insulation engineering and has successfully completed more than **750 projects** across industrial, commercial, and institutional sectors. The company has provided engineering consultancy services to organizations including Taj Hotels, Viacom18, Philips, Miraj Cinemas, Corona Remedies, CEPT University, Jindal Saw Ltd., Novotel, Gujarat Science City, Trützschler India, and ITC Hotels.

2.6 Leadership and Organizational Structure

Swa Consultancy was founded by **Mr. Viraj Shah**. The organization consists of a multidisciplinary team of senior experts, research engineers, and associate engineers who collaborate on design, simulation, technical documentation, site audits, and client engagement across its core service areas.

2.7 Internship Department

The internship was carried out within the **Research and Building Energy Modeling Division**, where the primary focus was on preparing CityBES-compatible datasets for Ahmedabad, integrating GIS-based building information with EnergyPlus simulations, and supporting Urban Building Energy Modeling (UBEM) studies for city-scale retrofit analysis.

3 Problem Statement

Buildings account for a significant share of global energy consumption and greenhouse gas emissions. With rapid urbanization and increasing energy demand, improving the energy performance of existing buildings has become an important step toward achieving sustainable urban development. Building energy retrofitting has emerged as an effective approach to reduce energy consumption, improve occupant comfort, and lower operational costs while minimizing environmental impacts.

Urban Building Energy Modeling (UBEM) enables the simulation of energy consumption for large groups of buildings by combining Geographic Information System (GIS) data, building characteristics, weather information, and building energy simulation engines such as EnergyPlus. These models allow planners, engineers, and policymakers to evaluate the energy performance of entire neighborhoods or cities, identify inefficient buildings, and assess the effectiveness of different retrofit strategies before their implementation.

City Building Energy Saver (CityBES) is a UBEM platform that facilitates city-scale energy simulations and supports the evaluation of various energy conservation measures. However, the successful application of such tools requires standardized and simulation-ready building datasets containing accurate geometric, physical, and operational information.

Therefore, there is a need to develop a systematic workflow for preparing CityBES-compatible building datasets for Ahmedabad. Such a workflow enables reliable building energy simulations, provides a foundation for evaluating retrofit strategies, and supports data-driven decision-making for sustainable urban planning and energy-efficient city development.

4 Objectives

Primary Objective

To develop a CityBES-compatible Urban Building Energy Modeling (UBEM) workflow for selected residential areas in Ahmedabad by preparing standardized GIS datasets, integrating climate-specific building parameters, performing EnergyPlus simulations, and evaluating the impact of energy-efficient retrofit strategies on building performance.

Secondary Objectives

- Extract residential building footprints from OpenStreetMap and preprocess the spatial datasets using QGIS.
- Estimate, validate, and incorporate building geometric attributes, including height, number of storeys, and building type.
- Prepare standardized GeoJSON datasets compatible with the CityBES platform for city-scale building energy modeling.
- Identify and integrate the appropriate EnergyPlus Weather (EPW) file representing Ahmedabad's climatic conditions.
- Incorporate Ahmedabad-specific operational parameters, HVAC characteristics, occupancy schedules, thermostat settings, and construction properties using Python scripting.
- Validate the prepared datasets and generate EnergyPlus simulation models through Energy Plus
- Perform baseline annual energy simulations to evaluate the existing energy performance of the selected residential districts.
- Implement Energy Conservation Measures (ECMs), including improvements to the building envelope, glazing system, cooling system efficiency, and lighting power density.
- Compare baseline and retrofit simulation results to quantify reductions in annual energy consumption, Energy Use Intensity (EUI), cooling demand, and operational carbon emissions.

5 Scope of the Project

The project focuses on developing an Urban Building Energy Modeling (UBEM) workflow for residential buildings located in the selected regions of Ahmedabad, namely Navrangpura and Thaltej. The scope includes spatial data collection, preprocessing, attribute generation, three-dimensional visualization, GeoJSON preparation, and integration of simulation-specific parameters required by the CityBES platform.

Building properties unavailable from publicly available datasets are estimated using standard engineering assumptions, literature, and geospatial analysis techniques. The prepared datasets are validated and converted into EnergyPlus-compatible simulation models to evaluate the baseline energy performance of the selected residential districts.

The scope further includes the implementation of Energy Conservation Measures (ECMs), such as improvements to the building envelope, glazing system, cooling system efficiency, and lighting power density, followed by retrofit energy simulations using EnergyPlus. The baseline and retrofit simulation results are compared to assess reductions in annual energy consumption, Energy Use Intensity (EUI), cooling demand, and operational carbon emissions. The developed workflow provides a scalable framework for city-scale building energy analysis, retrofit evaluation, and sustainable urban energy planning.

6 Literature Review / Background Study

Urban Building Energy Modeling (UBEM) has emerged as an important approach for evaluating the energy performance of large groups of buildings at the district and city scales. Unlike conventional building energy simulation, which focuses on individual buildings, UBEM integrates Geographic Information Systems (GIS), building characteristics, climate data, and energy simulation engines to estimate the energy consumption of thousands of buildings simultaneously.

Several software platforms have been developed for city-scale energy analysis, including EnergyPlus, OpenStudio, CityBES, and Urban Modeling Interface (UMI). Among these, City Building Energy Saver (CityBES) is an open-source web platform that utilizes the EnergyPlus simulation engine while providing an interactive GIS-based interface for district-level energy analysis and retrofit evaluation.

The primary input requirement for CityBES is a standardized GeoJSON dataset containing building geometries and essential building properties such as building height, number of storeys, building use type, construction year, and optional operational parameters through the `additional_json` field. Therefore, preparing accurate and validated GIS datasets becomes a critical step before energy simulations can be performed.

Previous studies have demonstrated that building geometry, operational schedules, HVAC systems, occupancy patterns, and climate-specific assumptions significantly influence building energy consumption. However, most default CityBES building prototypes are based on ASHRAE standards developed for North American buildings, which may not accurately represent Indian residential buildings. Consequently, incorporating Ahmedabad-specific operational assumptions such as cooling schedules, thermostat set-points, and HVAC configurations improves the reliability of simulation results.

During this internship, publicly available OpenStreetMap building footprint data were processed using QGIS to generate CityBES-compatible datasets. Additional parameters specific to Ahmedabad residential buildings were incorporated using Python scripting to improve the suitability of the datasets for city-scale energy simulation.

Existing Systems

The commonly used building energy modeling platforms include:

- EnergyPlus
- OpenStudio
- CityBES
- Urban Modeling Interface (UMI)

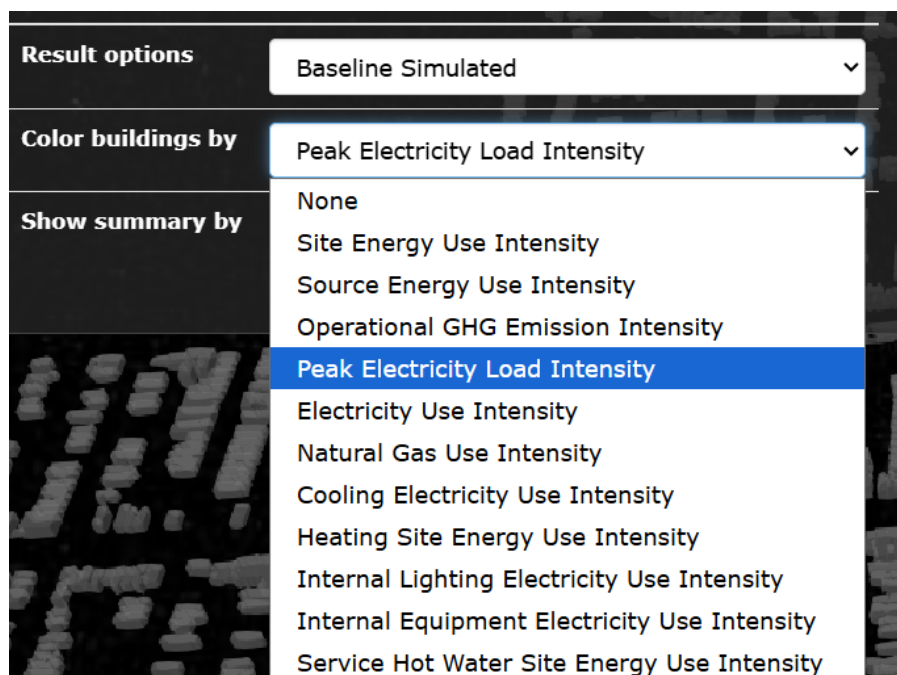


Figure 2: CityBES Colour Buildings by Features

Advantages of Existing Systems

- Accurate whole-building energy simulation
- Support for retrofit analysis
- Integration with GIS datasets
- Capability to analyze city-scale energy consumption
- Visualization of building energy performance

Challenges

- Require detailed building datasets
- Default assumptions are primarily based on international building standards
- Significant preprocessing of GIS data is required
- Preparation of standardized GeoJSON datasets can be time-consuming

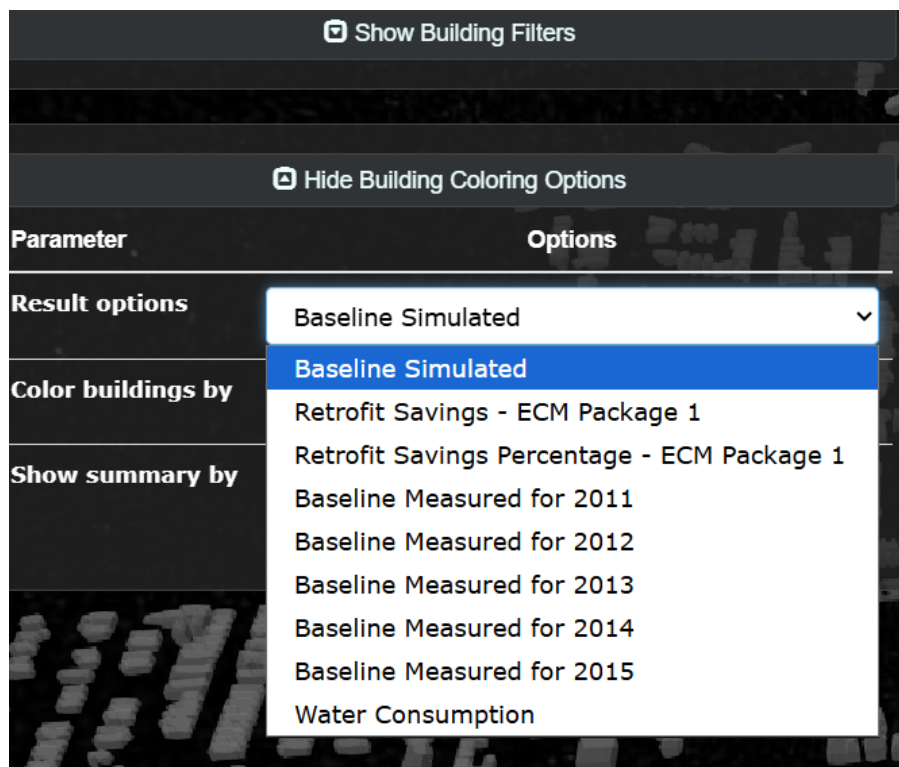


Figure 3: CityBES Result Options

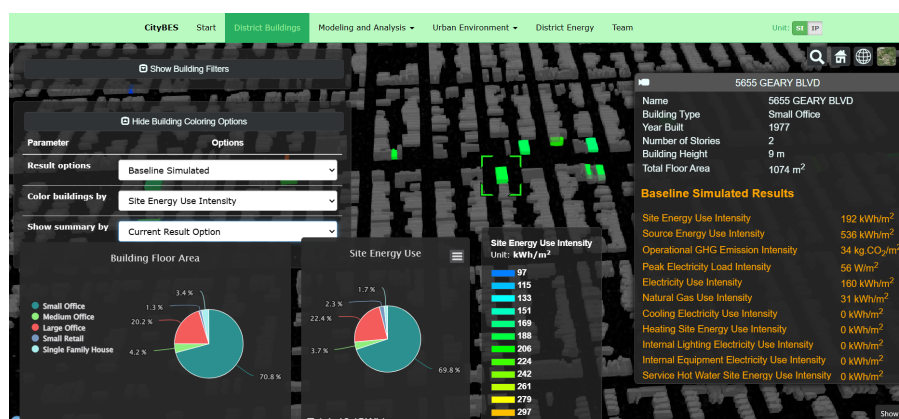


Figure 4: Baseline simulation results for San Francisco

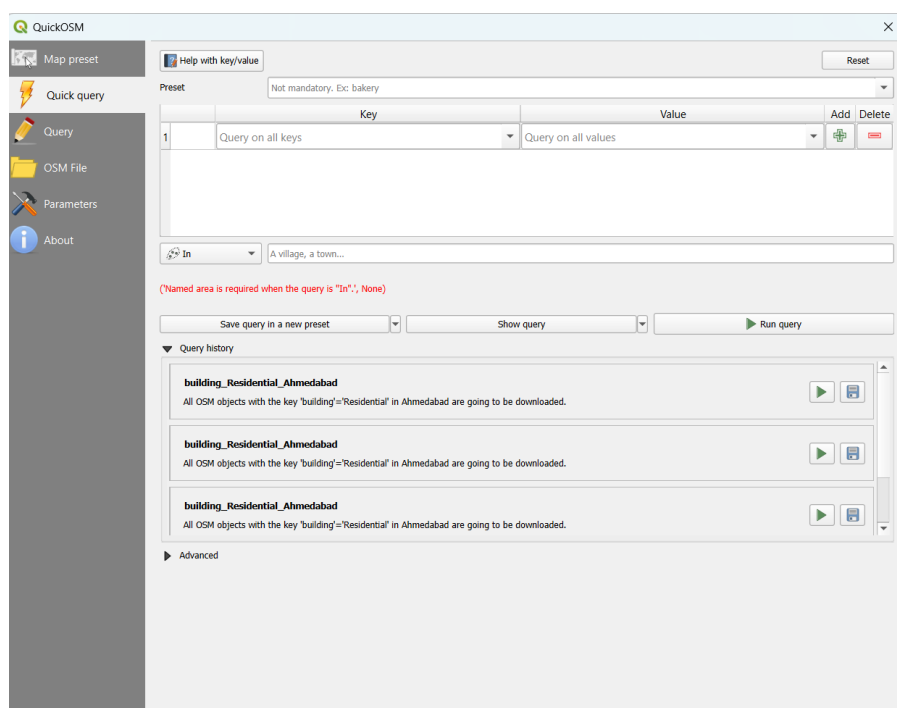


Figure 5: Quick OSM Plugin

7 Methodology

The project followed a systematic workflow for preparing UBE datasets compatible with the CityBES platform. The methodology included GIS data collection, selection of the appropriate Ahmedabad EnergyPlus Weather (EPW) file, spatial data preprocessing, building attribute generation, GeoJSON preparation, incorporation of Ahmedabad-specific operational parameters using Python, and preparation of simulation-ready datasets for CityBES.

Requirement Gathering

Initially, the project requirements were identified through discussions with the project mentor. The required inputs for CityBES, data formats, mandatory building attributes and simulation workflow were studied from the official CityBES documentation.

Data Collection

Initially, residential building datasets were explored for multiple regions of Ahmedabad, including Navrangpura, Thaltej, Vastrapur, Bopal, and Naroda, using OpenStreetMap through the QuickOSM plugin in QGIS. Building footprints and available attribute information such as building type, number of storeys, construction year, building material, roof material, and building colour were extracted wherever available.

The Ahmedabad 3D City Model was explored as a source for manually estimating

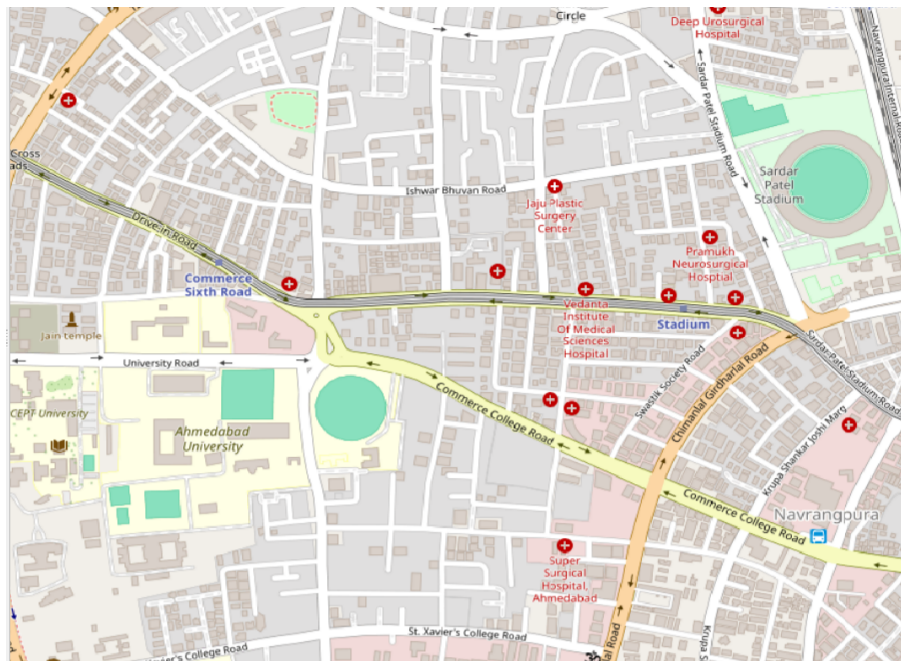


Figure 6: Open street map

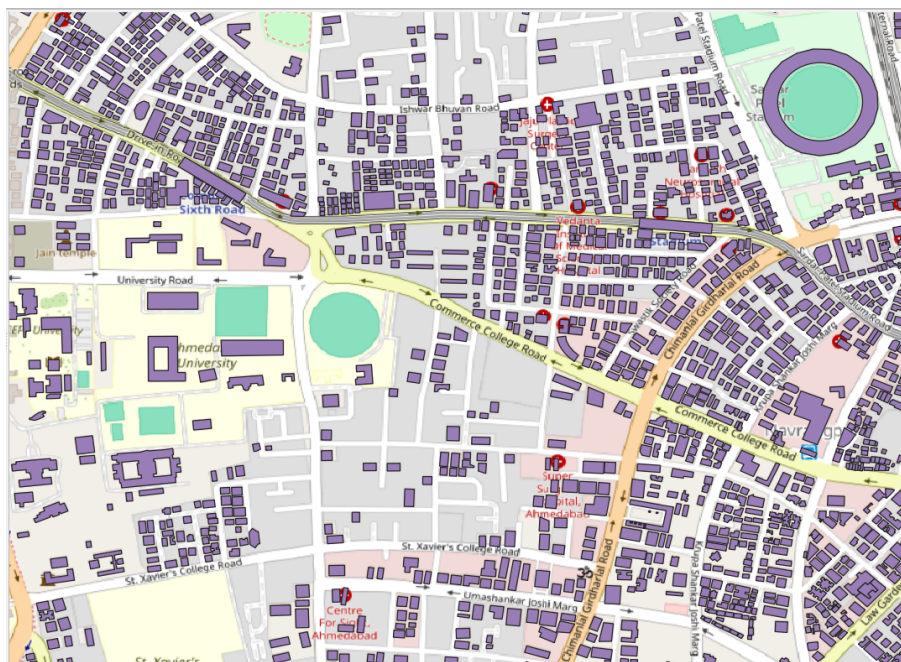


Figure 7: Building Foot Prints extracted

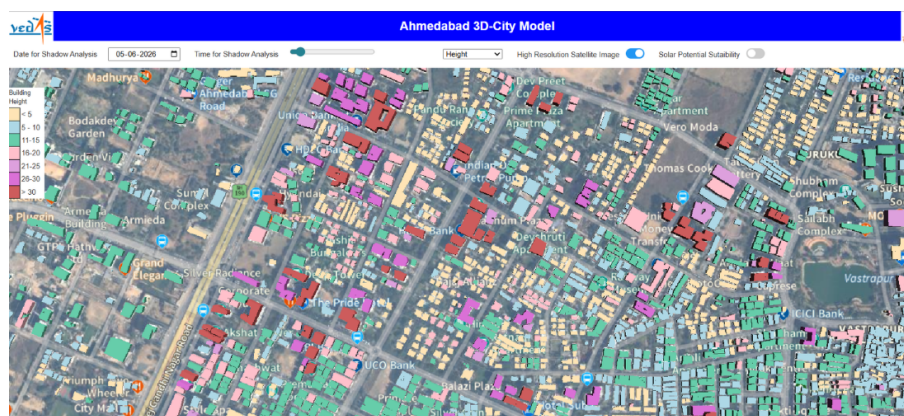


Figure 8: Ahmedabad 3D City Model

building heights. However, due to the large number of buildings and the absence of an efficient workflow for bulk data extraction, this approach was found to be impractical for city-scale dataset preparation.

Consequently, the study areas were finalized as Navrangpura and Thaltej, where sufficient **building:levels** information was available. Building heights were then estimated using the relationship:

$$BuildingHeight(m) = NumberOfStoreys \times 3$$

where an average floor-to-floor height of 3 m was assumed for residential buildings. This approach enabled the preparation of consistent and standardized datasets suitable for subsequent CityBES processing.

Data Preprocessing

The collected datasets were cleaned by removing unnecessary attributes and retaining only the fields required for building energy simulation. Missing building heights were estimated using available floor information based on standard floor-to-floor heights.

Additional geometric parameters including footprint area, total floor area, and building size classification were calculated using QGIS Field Calculator.

GeoJSON Preparation

The processed building datasets were exported as GeoJSON files while preserving the required coordinate reference system and geometry structure. The exported files were validated to ensure compatibility with CityBES.

Customization using Python

Python scripts were developed to automatically append the `additional_json` field to every building feature in the GeoJSON dataset. Ahmedabad-specific operational parameters including HVAC system type, cooling coefficient of performance (COP), thermostat setpoints, cooling schedules, heating schedules, and window properties were incorporated.

EPW Weather File Selection

An appropriate EnergyPlus Weather (EPW) file was required to provide the climatic inputs for building energy simulations in CityBES. Initially, the official EnergyPlus weather database was explored to obtain the weather file for Ahmedabad. However, the available files were based on older weather datasets and did not provide recently updated Typical Meteorological Year (TMY) data.

Alternative weather data repositories were then explored, including the Climate.OneBuilding repository, which provides EnergyPlus-compatible weather files generated using updated meteorological datasets. Multiple EPW files corresponding to Ahmedabad were reviewed and compared based on their temporal coverage and compatibility with CityBES.

After discussion with the project mentor, the file "`Weather_data_Ahmedabad.Intl.AP.426470.TMYx`" was selected for the project. This TMYx weather file represents typical meteorological conditions derived from the period 2009–2023 and provides hourly climatic parameters such as dry-bulb temperature, relative humidity, atmospheric pressure, wind speed, wind direction, direct and diffuse solar radiation, and precipitation. The selected EPW file was used as the climatic input for the baseline building energy simulations in CityBES.

8 Tools and Technologies Used

Softwares

- QGIS 3.40
- CityBES
- EnergyPlus
- Python 3.12
- Visual Studio Code
- Microsoft Excel
- Google Earth Pro

Programming Language

- Python

GIS Platform

- QGIS

Data Sources

- OpenStreetMap

Simulation Engine

- EnergyPlus

Version Control

- GitHub

Other Tools

- CityBES Web Platform
- GeoJSON
- Climate.OneBuilding

9 System Requirements

Hardware Requirements

Processor	Intel Core i5 or higher
RAM	Minimum 8 GB (16 GB Recommended)
Storage	Minimum 20 GB Free Space
Internet Connection	Required for CityBES and OpenStreetMap access
Graphics	Integrated or Dedicated Graphics supporting OpenGL

Software Requirements

Operating System	Windows 10/11 (64-bit)
GIS Software	QGIS 3.40
Programming Language	Python 3.12
Code Editor	Visual Studio Code
Simulation Platform	CityBES
Simulation Engine	EnergyPlus
Browser	Google Chrome / Microsoft Edge
Data Format	GeoJSON

10 Project Planning

Phase	Activity	Outcomes
Phase 1	Project Orientation and Background Study	Studied building energy fundamentals including heat transfer, thermal comfort, building envelope, insulation materials, energy retrofitting, Urban Building Energy Modeling (UBEM), building energy benchmarking, CityBES workflow, weather data requirements, and EnergyPlus simulation concepts.
Phase 2	GIS Data Collection and Processing	Extracted residential building footprints for Navrangpura and Thaltej using OpenStreetMap (QuickOSM plugin), explored additional geospatial data sources including VEDAS, Bhuvan, Google Earth Pro, Ahmedabad 3D City Model, and Copernicus Browser, and processed building attributes by calculating footprint area, total floor area, building heights, construction year, and other CityBES-compatible parameters.
Phase 3	Building Classification and GeoJSON Preparation	Classified buildings into Small, Medium, and Large categories using graduated symbology in QGIS, explored 3D visualization, exported CityBES-compatible GeoJSON files, reviewed GeoJSON structure, verified mandatory attributes, CRS requirements, geometry format, and mapped building attributes to CityBES input specifications.
Phase 4	Weather File Selection and Dataset Enhancement	Selected the Ahmedabad TMYx (2009–2023) EPW weather file after evaluating available weather databases. Developed Python scripts to automatically append Ahmedabad-specific <code>additional_json</code> parameters, including HVAC configuration, thermostat settings, operational schedules, window properties, and other simulation inputs.
Phase 5	CityBES Dataset Preparation and Baseline Simulation	Validated the prepared GeoJSON datasets, uploaded them to the CityBES platform, generated EnergyPlus simulation models, and performed baseline annual energy simulations to evaluate the existing energy performance of the residential building stock.
Phase 6	Retrofit Strategy Development	Identified suitable Energy Conservation Measures (ECMs), including external Polyurethane Foam (PUF) insulation, Vacuum Insulated Glazing (VIG), improved cooling system efficiency, and reduced lighting power density. Calculated construction assembly U-values and prepared baseline and retrofit construction assemblies for simulation.
Phase 7	Retrofit Simulation and Performance Evaluation	Implemented the selected retrofit measures in CityBES/EnergyPlus and performed annual retrofit simulations for Navrangpura and Thaltej. Compared baseline and retrofit models by evaluating annual site energy, source energy, Energy Use Intensity (EUI), cooling energy, and operational carbon emissions.
Phase 8	Result Analysis and Documentation	Analyzed simulation results using comparative tables and graphical visualization, quantified the energy savings achieved through retrofit implementation, and prepared the final internship report, presentation, and technical documentation.

Table 1: Project Planning Timeline

11 Implementation

The implementation phase involved converting publicly available GIS building data into simulation-ready datasets compatible with the CityBES platform. The workflow consisted of multiple processing stages, beginning with spatial data collection and ending with preparation of customized GeoJSON files for energy simulation.

11.1 Module 1: Data Collection

Residential building footprints for the selected study areas, Navrangpura and Thaltej, Ahmedabad, were extracted from OpenStreetMap using the QuickOSM plugin in QGIS. Building polygons and available attribute information such as building type, number of storeys, construction year, roof material and building material were collected.

Output:

- Residential building footprint dataset
- Building attribute table
- Geographic boundary of study areas

11.2 Module 2: Data Cleaning and Attribute Preparation

The extracted datasets contained numerous unnecessary OpenStreetMap attributes. The attribute tables were cleaned by removing irrelevant columns while retaining only the fields required for building energy simulation.

Additional parameters including footprint area, building height, total floor area, and building size classification were generated using the QGIS Field Calculator.

Building heights were estimated from the number of storeys using a floor-to-floor height assumption of approximately 3 m.

Generated attributes include:

- Building Height
- Building Area
- Total Floor Area
- Building Size Class
- Estimated Construction Year

full_id	osm_id	building	building:levels	building:colour	Area	longitude	latitude	Height	built_year	total_floor_area	size_class
w317502589	317502589	apartments	7	lightyellow	497.087	72.557073	23.025458	21	1998	3479.609	Large
w317629868	317629868	apartments	10	orange	915.023	72.557242	23.027918	30	2019	9150.23	Large
w317658411	317658411	residential	2	white	179.217	72.555601	23.039213	6	2024	358.434	Small
w317658412	317658412	residential	2	white	245.518	72.556083	23.039202	6	2009	491.036	Small
w317658414	317658414	residential	2	white	212.777	72.55585	23.039206	6	2004	425.554	Small
w317658416	317658416	residential	2	white	273.727	72.555364	23.039381	6	2017	547.454	Small
w324736856	324736856	residential	10	NULL	534.104	72.562423	23.030721	30	1993	5341.04	Large
w324736857	324736857	residential	10	NULL	444.215	72.562881	23.030686	30	2002	4442.15	Large
w326192661	326192661	residential	3	white	194.727	72.565681	23.034832	9	2005	584.181	Small
w326196542	326196542	apartments	5	white	461.493	72.566986	23.032585	15	2023	2307.465	Medium
w326196545	326196545	apartments	5	white	175.689	72.566884	23.032344	15	1993	878.445	Medium
w326196547	326196547	apartments	5	white	428.951	72.566753	23.03246	15	1986	2144.755	Medium
w326196548	326196548	apartments	7	white:prange	215.714	72.567117	23.032171	21	2002	1509.998	Medium
w327199740	327199740	residential	4	beige	2025.313	72.56992	23.02972	12	1997	8101.252	Large
w327199750	327199750	residential	10	beige	799.594	72.569211	23.029802	30	2006	7995.94	Large
w368910606	368910606	apartments	5	whitered	485.3	72.565219	23.034112	15	1989	2426.5	Medium
w368910607	368910607	apartments	5	whitered	437.511	72.565367	23.034	15	2016	2187.555	Medium
w368910608	368910608	apartments	5	white	357.938	72.565522	23.03388	15	2000	1789.69	Medium
w376971936	376971936	residential	2	beige	81.471	72.549229	23.041195	6	1991	162.942	Small
w376971937	376971937	residential	2	beige	273.731	72.549294	23.041619	6	1991	547.462	Small
w376971940	376971940	residential	2	beige	83.454	72.549234	23.041272	6	2024	166.908	Small
w376971941	376971941	residential	2	white	60.065	72.549112	23.040438	6	1985	120.13	Small
w376971942	376971942	residential	2	beige	173.016	72.549303	23.040789	6	2004	346.032	Small

Figure 9: Attribute table

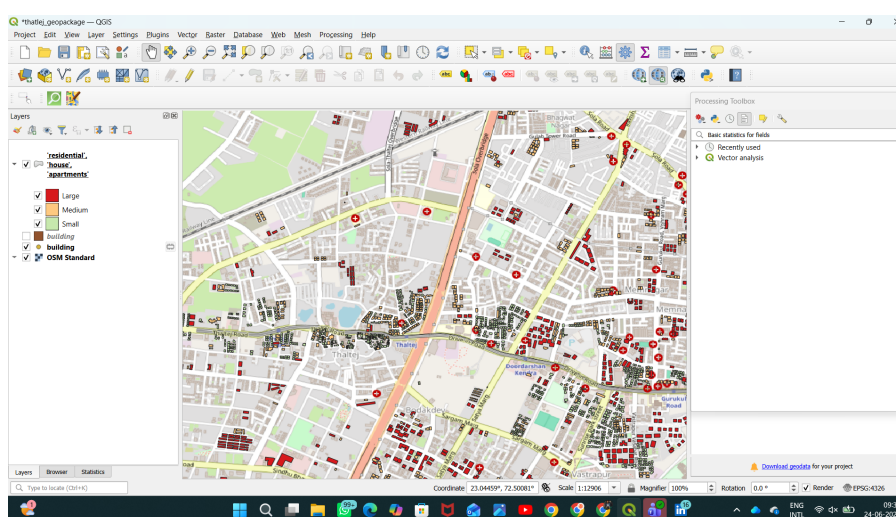


Figure 10: Buildings classified based on area

11.3 Module 3: Building Classification and Visualization

After calculating the footprint area of each building, the residential buildings were classified into three categories based on their size:

- Small Buildings (Green)
- Medium Buildings (Yellow)
- Large Buildings (Red)

The classification was performed using graduated symbology in QGIS, where different colors were assigned to each category for better visualization and verification of the processed datasets. This helped in identifying the spatial distribution of buildings of different sizes and validating the calculated building areas before exporting the final GeoJSON datasets.

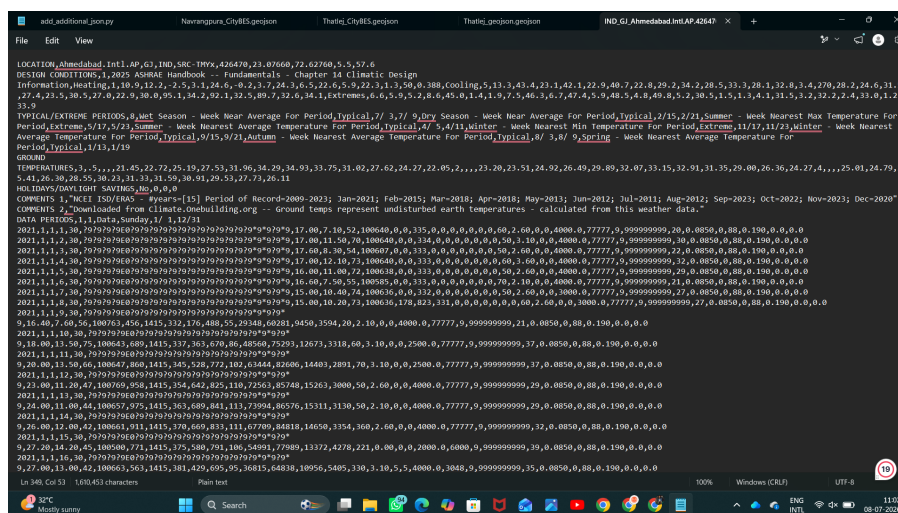


Figure 11: Weather file

Additionally, the feasibility of generating three-dimensional building models using the QGIS 3D Map View was briefly explored through attribute-based extrusion. This was carried out as an exploratory exercise to understand QGIS visualization capabilities and was not included in the final CityBES dataset preparation workflow.

11.4 Module 4: EPW Weather File Selection

An appropriate EnergyPlus Weather (EPW) file was required to provide the climatic input for building energy simulations in CityBES. Initially, the official EnergyPlus weather database was explored to obtain a suitable weather file for Ahmedabad. However, the available files were based on older meteorological datasets and did not include recently updated Typical Meteorological Year (TMY) weather data.

Alternative weather data repositories were subsequently explored, including the Climate.OneBuilding repository, which provides EnergyPlus-compatible weather files generated using updated climatic datasets. Multiple EPW files corresponding to Ahmedabad were reviewed and compared based on their temporal coverage, weather data source, and compatibility with CityBES.

After discussion with the project mentor, the file `Weather_data_Ahmedabad.Int1.AP.426470.TMYx.1` was selected for the project. This TMYx weather file represents typical meteorological conditions derived from the period 2009–2023 and provides hourly climatic parameters such as dry-bulb temperature, relative humidity, wind speed, wind direction, atmospheric pressure, and direct and diffuse solar radiation. The selected EPW file serves as the climatic boundary condition for baseline building energy simulations in CityBES.

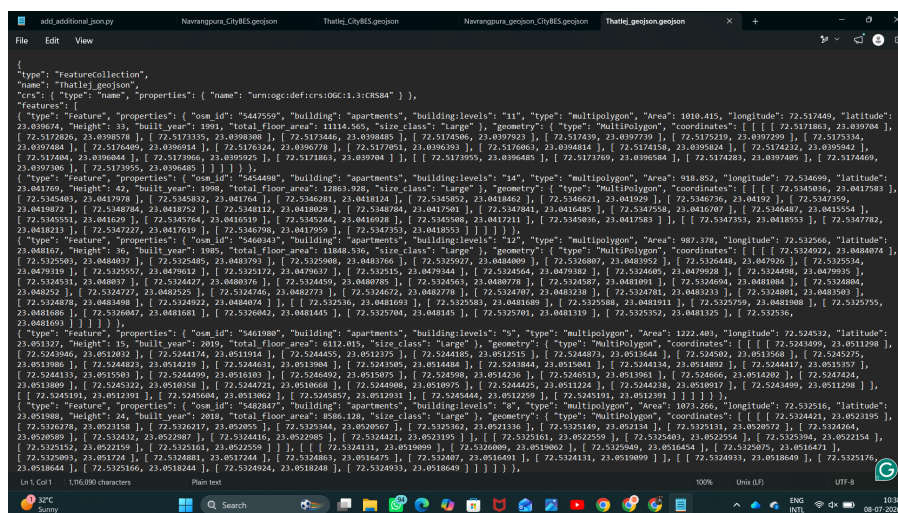


Figure 12: Geojson code

11.5 Module 5: GeoJSON Generation

After preprocessing, the finalized building datasets were exported as GeoJSON files. The exported files preserved both the polygon geometry and associated building attributes.

The generated GeoJSON structure consisted of:

- FeatureCollection
- Feature objects
- Building properties
- Polygon geometry
- Geographic coordinates

The exported files were manually inspected to ensure compliance with the CityBES input specifications.

11.6 Module 6: Python-Based Dataset Enhancement

A Python script was developed to automate the modification of GeoJSON datasets.

The script iterated through every building feature and appended an `additional_json` object containing Ahmedabad-specific operational parameters.

The customized parameters incorporated into the `additional_json` field included:

- HVAC system type (`hvac_system_type_id`)
- HVAC cooling coefficient of performance (COP)
- Heating thermostat setpoint

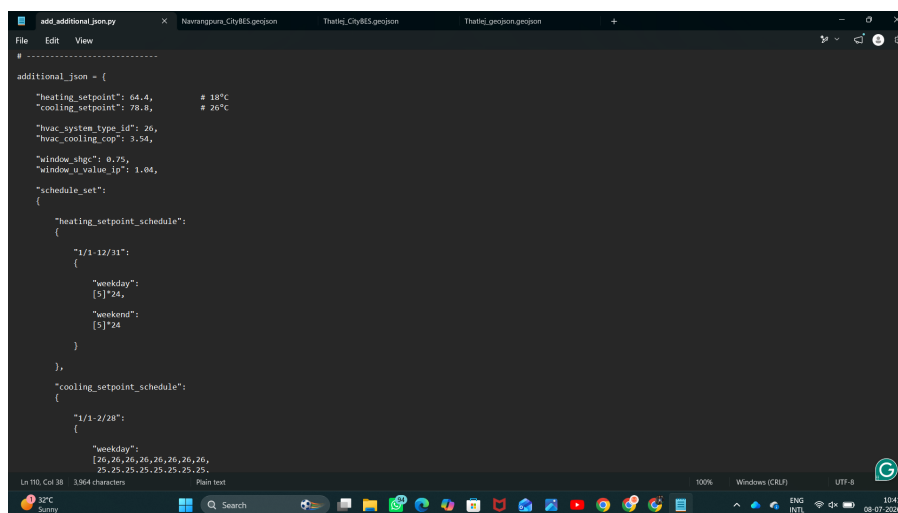


Figure 13: Additional Json code

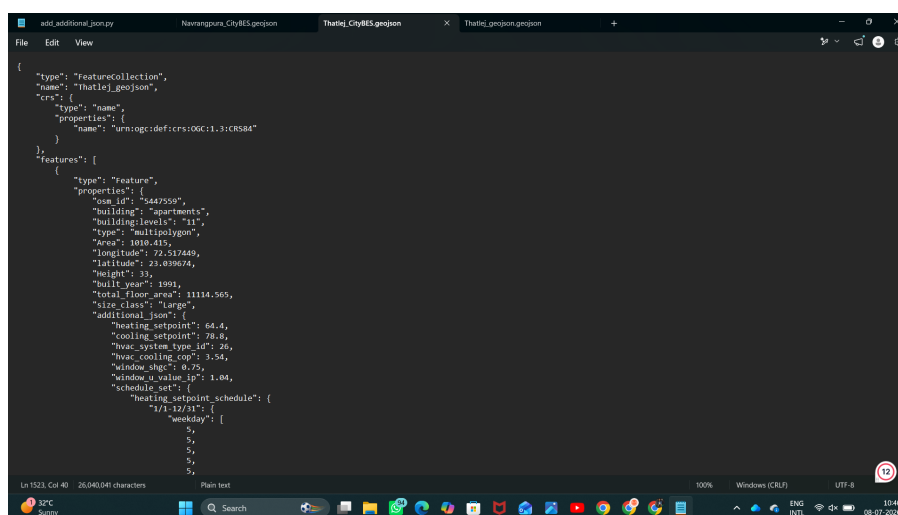


Figure 14: After integrating additional json code

- Cooling thermostat setpoint
- Window Solar Heat Gain Coefficient (SHGC)
- Window U-value
- Heating setpoint schedule
- Cooling setpoint schedule

All other operational parameters were retained as the default CityBES prototype values. The customized parameters were selected to better represent Ahmedabad residential buildings based on the project requirements and mentor recommendations.

Automating this process ensured consistency across all buildings and eliminated manual editing errors.

11.7 Module 7: CityBES Compatibility Check

The prepared GeoJSON datasets were reviewed to ensure that they conformed to the input requirements specified in the CityBES data preparation guidelines. The following aspects were checked before preparing the datasets for upload:

- GeoJSON file structure and syntax
- Presence of mandatory building attributes
- Coordinate Reference System (CRS) - EPSG:4326 — WGS 84
- Building height units (metres)
- Ahmedabad-specific `additional_json` parameters
- Operational schedules
- Overall compatibility with the CityBES input format

The finalized GeoJSON datasets were then prepared for subsequent upload and baseline energy simulation using the CityBES platform.

11.8 Module 8: EnergyPlus Simulation and Performance Analysis

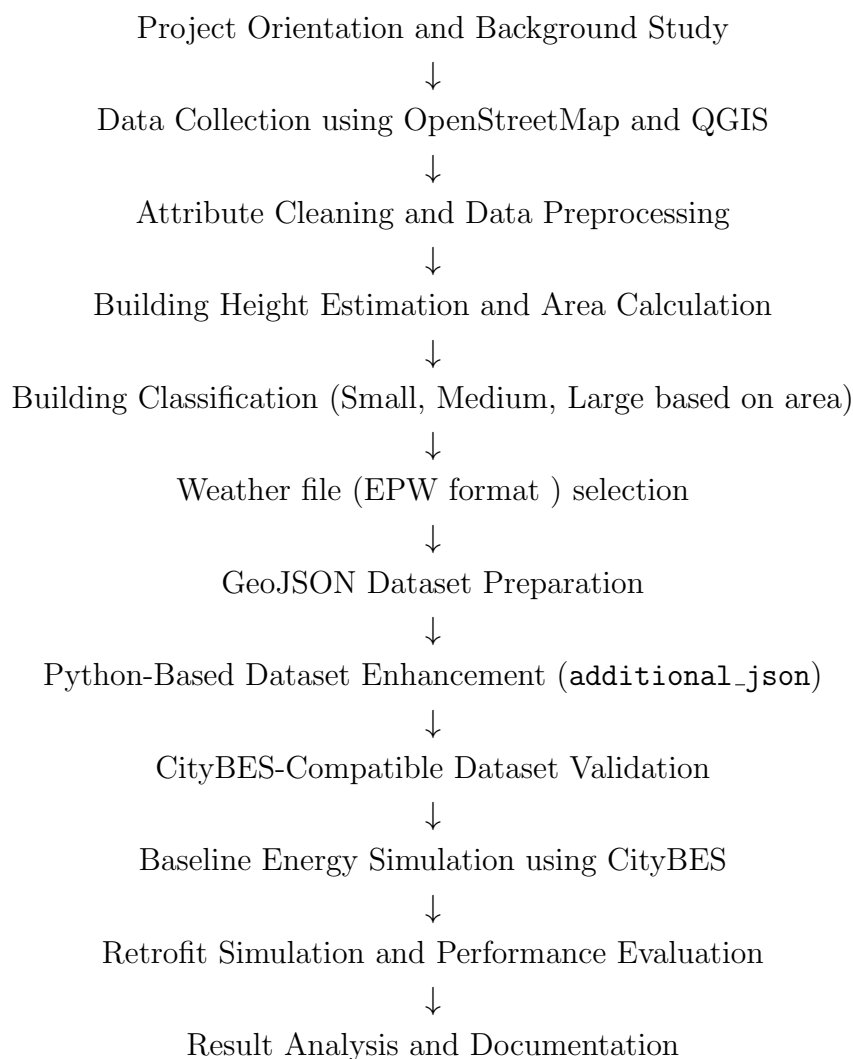
After verifying the prepared datasets, the corresponding EnergyPlus simulation models were generated and annual building energy simulations were performed using the selected Ahmedabad Typical Meteorological Year (TMYx) weather file. Baseline simulations were initially conducted to evaluate the existing energy performance of the residential buildings.

Following the baseline analysis, retrofit simulation models were prepared by modifying the required building and system parameters. Annual EnergyPlus simulations were then executed under identical climatic conditions to evaluate the impact of the proposed retrofit strategy.

The simulation outputs were analyzed by comparing the baseline and retrofit models using key building energy performance indicators, including annual site energy, source energy, Energy Use Intensity (EUI), end-use electricity consumption, and operational carbon emissions. The results were interpreted using comparative tables and graphical visualizations to assess the effectiveness of the retrofit strategy for the selected residential districts.

11.9 Workflow

The overall implementation workflow adopted during the internship is illustrated below:



11.10 Algorithms and Processing Techniques

The project primarily involved GIS processing and data transformation rather than conventional algorithm development. The following computational techniques were employed:

- Spatial Querying
- Polygon Geometry Processing
- Attribute-Based Classification
- Rule-Based Height Estimation
- Field Calculations in QGIS

- JSON Parsing
- Python Dictionary Manipulation
- Automated GeoJSON Modification

12 Testing and Validation

Several validation procedures were carried out to ensure that the prepared datasets were compatible with the CityBES platform and suitable for Urban Building Energy Modeling. The generated GeoJSON datasets were compared with the example dataset provided in the CityBES documentation. Field names, geometry structure, data types, and additional json code syntax were verified to ensure compatibility before upload.

12.1 Functional Testing

Each processing stage was verified independently

- Correct extraction of residential buildings
- Successful calculation of geometric attributes
- Proper generation of building heights
- Accurate classification of buildings
- Successful export to GeoJSON format

12.2 Data Check

The generated GeoJSON files were validated against the CityBES data preparation guidelines

The following checks were performed:

- Presence of mandatory fields
- Correct GeoJSON syntax
- Valid polygon geometry
- Height values expressed in metres
- Correct coordinate reference system
- Proper formatting of the `additional_json` object

12.3 Script Check

The Python automation script was tested to ensure that:

- Every building feature received the `additional_json` object
- No existing building attributes were modified unintentionally
- Output GeoJSON files remained syntactically valid

13 Results and Discussion

13.1 Baseline Energy Simulations

After preparing the CityBES-compatible GeoJSON datasets and converting them into EnergyPlus IDF models, baseline annual energy simulations were performed using EnergyPlus with the selected Ahmedabad Typical Meteorological Year (TMYx) weather file.

The objective of the baseline simulation was to establish the existing energy performance of the residential building stock before evaluating any retrofit measures. The generated baseline results serve as the reference case for comparing future energy-saving strategies.

Parameter	Navrangpura	Thaltej
Total Building Area (m ²)	825269.59	1191137.13
Conditioned Area (m ²)	412634.79	595568.57
Total Site Energy (GJ)	429048.28	651649.56
Total Source Energy (GJ)	1358795.89	2063774.14
Site Energy Intensity (MJ/m ²)	519.89	547.08
Source Energy Intensity (MJ/m ²)	1646.49	1732.61

Table 2: Baseline Energy Simulation Summary

The baseline simulations indicated that cooling, interior lighting, and interior equipment constituted the major components of annual electricity consumption in both study areas. Heating energy was negligible because Ahmedabad experiences a predominantly hot climate and residential buildings generally do not require space heating throughout most of the year.

The results therefore indicate that cooling-related retrofit measures have the greatest potential for reducing annual building energy consumption.

End Use	Navrangpura (GJ)	Thaltej (GJ)
Cooling	139588.60	233170.63
Interior Lighting	156154.21	225382.20
Interior Equipment	130128.51	187818.50
Fans	3176.96	5278.22
Heating	0.00	0.00

Table 3: Major Annual End Uses

Although Thaltej consumed more total energy than Navrangpura, this was expected because of its larger building stock and total floor area. When normalized using energy intensity (MJ/m²), both study areas demonstrated comparable baseline performance, indicating consistency in the developed workflow and simulation assumptions.

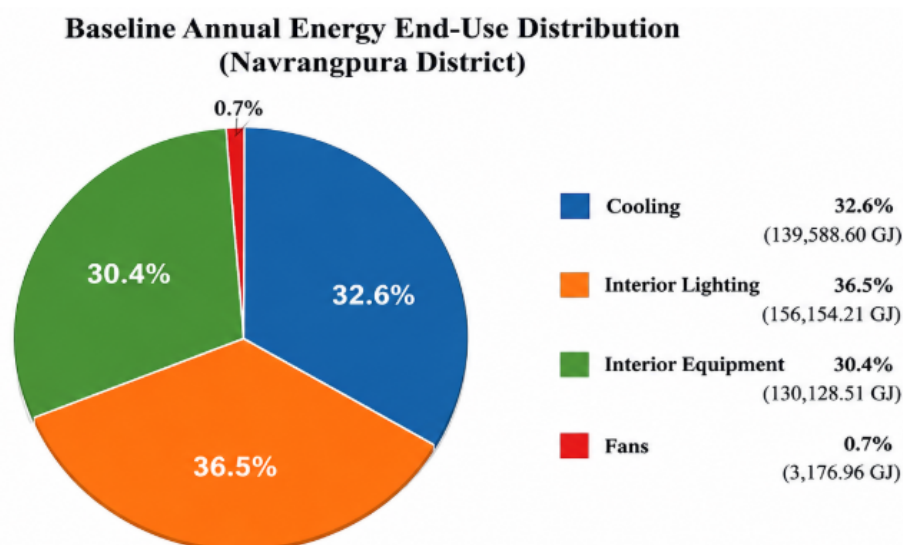


Figure 13.1: Baseline annual energy end-use distribution for Navrangpura District
(Values in bracket indicate energy consumption in GJ).

Figure 13.1 illustrates the distribution of annual site energy consumption among the major end uses in the Navrangpura residential building stock. Interior lighting accounts for the largest share of the total energy consumption (36.5%), followed by cooling (32.6%) and interior equipment (30.4%). Fan energy constitutes only 0.7% of the total annual site energy, while space heating is negligible due to Ahmedabad's predominantly hot climate.

The results indicate that nearly all annual energy consumption is associated with cooling and electrical end uses. This highlights that improving cooling system efficiency, adopting energy-efficient lighting technologies, and reducing internal equipment loads have the greatest potential for lowering the overall energy consumption of the residential buildings in Navrangpura.

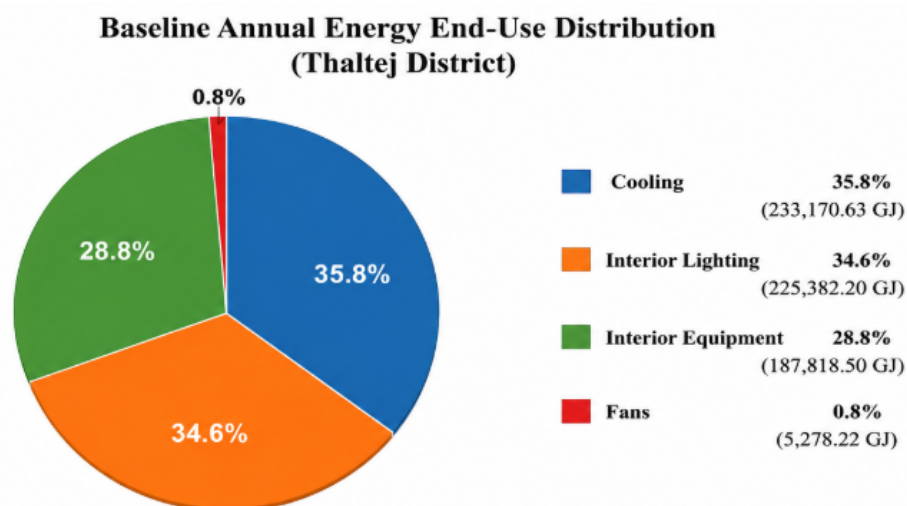


Figure 13.2: Baseline annual energy end-use distribution for Thaltej District
(Values in bracket indicate energy consumption in GJ).

Figure 13.2 presents the annual site energy end-use distribution for the Thaltej residential building stock. Cooling represents the largest component of annual energy consumption (35.8%), followed closely by interior lighting (34.6%) and interior equipment (28.8%). Fan energy contributes only 0.8% of the total site energy, whereas heating energy remains negligible because of Ahmedabad's cooling-dominated climate.

Compared with Navrangpura, Thaltej exhibits a slightly higher proportion of cooling energy demand, indicating a greater dependence on mechanical cooling systems. The baseline simulation therefore suggests that cooling-related retrofit strategies, together with improvements in lighting efficiency and equipment performance, are expected to provide the most significant reductions in annual building energy consumption.

Overall, the baseline simulations successfully established the existing energy performance of the residential building stock in both study areas. The simulations demonstrate that cooling, interior lighting, and equipment loads dominate the annual energy demand, while heating energy is practically insignificant under Ahmedabad's climatic conditions. These baseline results provide the reference against which the effectiveness of the proposed retrofit strategies will be evaluated in the subsequent section.

13.2 Evaluation of Retrofit Strategies

13.2.1 Retrofitting

Buildings account for a major share of global energy consumption, with a significant portion of this energy being used for maintaining indoor thermal comfort. Many existing residential buildings were constructed without adequate thermal insulation or energy-efficient building systems, resulting in high cooling demands, especially in hot climates such as Ahmedabad.

Building retrofitting refers to the process of upgrading existing buildings by improving the building envelope, mechanical systems, and operational characteristics to enhance energy performance while preserving the original structure. Rather than demolishing and reconstructing buildings, retrofitting provides a cost-effective and environmentally sustainable approach for reducing energy consumption, lowering greenhouse gas emissions, and improving occupant comfort.

Energy retrofit measures may include improvements to wall and roof insulation, replacement of inefficient glazing systems, enhancement of HVAC system efficiency, installation of energy-efficient lighting, and optimization of building operational schedules. These measures reduce heat transfer through the building envelope, decrease cooling loads, and improve the overall energy efficiency of the building.

In this project, retrofit strategies were evaluated using the EnergyPlus simulation engine. The baseline building models prepared for Navrangpura and Thaltej were modified by incorporating selected Energy Conservation Measures (ECMs), and the resulting energy performance was compared with the baseline simulations to quantify the potential energy savings achievable for residential buildings in Ahmedabad.

13.2.2 Energy Conservation Measures (ECMs)

Energy Conservation Measures (ECMs) are retrofit strategies implemented to reduce the energy consumption of buildings while maintaining or improving occupant comfort. In Urban Building Energy Modeling (UBEM), ECMs are evaluated by modifying building envelope characteristics, HVAC systems, lighting systems, operational schedules, and other building parameters to estimate their impact on annual energy performance.

CityBES provides a library of predefined ECMs that can be applied individually or in combination to analyze potential energy savings at the building and district scales. These measures are based on modifications to building components and system efficiencies and are simulated using the EnergyPlus simulation engine.

The major categories of ECMs available in CityBES include:

1. Roof insulation improvement
2. Exterior wall insulation improvement

3. High-performance window replacement
4. Window-to-wall ratio modification
5. HVAC system efficiency improvement
6. Lighting power density reduction
7. Service hot water system improvement

Each ECM influences a different aspect of building energy performance. Envelope-related measures reduce heat transfer through the building envelope, HVAC upgrades improve system efficiency, and lighting improvements decrease internal electrical loads. The effectiveness of these measures depends on the local climate, building characteristics, occupancy patterns, and operational schedules.

13.2.3 ECMs Implemented in the Present Study

Although CityBES provides several Energy Conservation Measures (ECMs) for improving building energy performance, this study focused on retrofit strategies that are practical and suitable for existing residential buildings in Ahmedabad. The selected measures primarily targeted the building envelope and cooling system, which have the greatest influence on energy consumption in cooling-dominated climates.

The following retrofit measures were implemented in the EnergyPlus simulations:

- Addition of external Polyurethane Foam (PUF) insulation to exterior walls.
- Addition of external PUF insulation over the roof slab.
- Replacement of existing single-glazed windows with Vacuum Insulated Glazing (VIG) windows having lower U-value and Solar Heat Gain Coefficient (SHGC).
- Improvement of HVAC system efficiency by increasing the cooling Coefficient of Performance (COP) from 3.8 to 4.5.
- Reduction of Lighting Power Density (LPD) from 6 W/m² to 5 W/m² through the adoption of energy-efficient lighting systems.

The thermostat setpoints, occupancy schedules, operational schedules, and other building operational parameters were kept unchanged from the baseline model. This ensured that the observed reduction in energy consumption was solely due to the implemented retrofit measures, allowing a fair comparison between the baseline and retrofit scenarios.

13.2.4 Polyurethane Foam (PUF) Insulation and External Insulation Strategy

Polyurethane Foam (PUF) insulation was selected as the primary retrofit material for the building envelope due to its excellent thermal performance and suitability for retrofitting existing buildings. PUF possesses one of the lowest thermal conductivities among commercially available insulation materials ($\lambda \approx 0.024 \text{ W/m}\cdot\text{K}$), enabling significant reductions in heat transfer with relatively small insulation thicknesses. This allows the retrofit to achieve substantial improvements in thermal resistance without requiring major alterations to the existing building structure.

In addition to its high insulating capability, PUF offers several practical advantages for retrofit applications. The material is lightweight, minimizing additional structural loads on existing buildings, and its closed-cell structure provides very low moisture absorption, helping maintain long-term thermal performance under varying climatic conditions. PUF insulation boards are also easy to transport, cut and install, thereby reducing construction time and minimizing disruption to building occupants during renovation works.

For the present study, the insulation was applied as an **external insulation system**. External insulation creates a continuous thermal envelope around the building, thereby significantly reducing thermal bridges at slabs, beams and columns. Unlike internal insulation, it preserves usable indoor floor area and can generally be installed without requiring occupants to vacate the building. Furthermore, the existing masonry walls remain within the insulated envelope, increasing their thermal mass and improving indoor thermal stability by reducing temperature fluctuations.

Previous studies have demonstrated that external insulation systems are among the most effective retrofit measures for reducing cooling energy demand in existing buildings, particularly in hot climatic regions similar to Ahmedabad. The combination of high-performance PUF insulation with an external insulation strategy therefore provides an energy-efficient and practical solution for improving the thermal performance of the selected building stock.

13.2.5 U-Value Calculations

The thermal transmittance (U-value) of each building envelope component was calculated in accordance with ISO 6946, which determines the overall heat transfer coefficient by considering the thermal resistance of each material layer together with the internal and external surface resistances. The total thermal resistance is given by

$$R_{total} = R_{si} + \sum_{i=1}^n \frac{t_i}{k_i} + R_{se} \quad (1)$$

where:

- R_{si} = Internal surface resistance = $0.13 \text{ m}^2\text{K/W}$

- R_{se} = External surface resistance = 0.04 m²K/W
- t_i = Thickness of material layer (m)
- k_i = Thermal conductivity of material (W/m·K)

The overall thermal transmittance is then calculated as

$$U = \frac{1}{R_{total}} \quad (2)$$

The calculations for the baseline and retrofit assemblies are presented below.

Baseline Wall

$$R_{wall} = \frac{0.200}{0.324} = 0.617 \text{ m}^2\text{K/W}$$

$$R_{total} = 0.617 + 0.13 + 0.04 = 0.787 \text{ m}^2\text{K/W}$$

$$U = \frac{1}{0.787} = 1.27 \text{ W/m}^2\text{K}$$

Baseline Roof

$$R_{roof} = \frac{0.150}{0.213} = 0.704 \text{ m}^2\text{K/W}$$

$$R_{total} = 0.704 + 0.13 + 0.04 = 0.874 \text{ m}^2\text{K/W}$$

$$U = \frac{1}{0.874} = 1.14 \text{ W/m}^2\text{K}$$

Retrofit Wall The retrofit wall consists of a 60 mm PUF insulation layer applied externally over the existing 200 mm masonry wall.

$$R_{PUF} = \frac{0.060}{0.024} = 2.50 \text{ m}^2\text{K/W}$$

$$R_{wall} = \frac{0.200}{0.324} = 0.617 \text{ m}^2\text{K/W}$$

$$R_{total} = 2.50 + 0.617 + 0.13 + 0.04 = 3.287 \text{ m}^2\text{K/W}$$

$$U = \frac{1}{3.287} = 0.30 \text{ W/m}^2\text{K}$$

Retrofit Roof The retrofit roof consists of a 65 mm PUF insulation layer applied externally over the existing RCC slab.

$$R_{PUF} = \frac{0.065}{0.024} = 2.708 \text{ m}^2\text{K/W}$$

$$R_{RCC} = \frac{0.150}{0.213} = 0.704 \text{ m}^2\text{K/W}$$

$$R_{total} = 2.708 + 0.704 + 0.13 + 0.04 = 3.582 \text{ m}^2\text{K/W}$$

$$U = \frac{1}{3.582} = 0.279 \approx 0.28 \text{ W/m}^2\text{K}$$

The retrofit strategy reduced the wall U-value from 1.27 to 0.30 W/m²K (approximately 76% reduction) and the roof U-value from 1.14 to 0.28 W/m²K (approximately 75% reduction). These reductions significantly decrease conductive heat transfer through the building envelope, thereby lowering cooling loads and improving overall energy efficiency.

13.2.6 Construction Assemblies for Baseline and Retrofit

The construction assemblies used in the EnergyPlus simulations are summarized in this section. The baseline model represents the existing building envelope commonly found in residential buildings in Ahmedabad, consisting of masonry walls, reinforced cement concrete (RCC) roofs, and single-glazed windows. The retrofit model incorporates external Polyurethane Foam (PUF) insulation for the opaque envelope and Vacuum Insulated Glazing (VIG) for fenestration to improve the overall thermal performance of the building envelope. The calculated assembly U-values obtained in the previous subsection were assigned to the corresponding construction assemblies in the simulation model.

Construction	Layers (Outside → Inside)	Thickness (mm)	Conductivity (W/m·K)	Assembly U-value (W/m ² K)
Wall	Masonry	200	0.324	1.27
Roof	RCC Slab	150	0.213	1.14
Window	Single Glazing	–	U = 5.905, SHGC = 0.75	5.905

Table 4: Baseline Construction Assemblies

Construction	Layers (Outside → Inside)	Thickness (mm)	Conductivity (W/m·K)	Assembly U-value (W/m ² K)
Wall	PUF Insulation + Masonry	60 + 200	0.024 + 0.324	0.30
Roof	PUF Insulation + RCC Slab	65 + 150	0.024 + 0.213	0.28
Window	Vacuum Insulated Glazing (VIG)	–	U = 0.60, SHGC = 0.287	0.60

Table 5: Retrofit Construction Assemblies

Parameter	Baseline	Retrofit	Improvement
Cooling COP (DX Coil)	3.8	4.5	+18%
Lighting Power Density (W/m ²)	6	5	–17%

Table 6: System Parameters Used in Baseline and Retrofit Models

The retrofit construction assemblies substantially improve the thermal resistance of the building envelope by reducing the U-values of walls, roofs and windows. Together with improvements in cooling system efficiency and lighting power density, these modifications form the input parameters for the retrofit EnergyPlus simulations presented in the following sections.

13.2.7 Retrofit Simulation Results

The retrofit building models developed for the Navrangpura and Thaltej districts were simulated using EnergyPlus through the CityBES platform under Ahmedabad Typical Meteorological Year (TMYx) weather conditions. The retrofit strategies described in the previous subsections were incorporated into the simulation models, including improvements to the building envelope, window systems, cooling equipment, and lighting efficiency.

The simulation outputs provide annual building utility performance and end-use energy consumption for each district. The results presented in this section represent the annual energy performance after implementing the selected retrofit measures.

13.2.8 Navrangpura District

Table 13.2.8 summarizes the annual building utility performance obtained after applying the selected retrofit measures to the Navrangpura district model.

Parameter	Value
Total Building Area (m ²)	825269.59
Conditioned Area (m ²)	412634.79
Total Site Energy (GJ)	359939.78
Site Energy Intensity (MJ/m ²)	436.15
Total Source Energy (GJ)	1139929.29
Source Energy Intensity (MJ/m ²)	1381.28

Table 7: Annual Energy Performance of Navrangpura Retrofit Model

End Use	Energy (GJ)
Cooling	97069.89
Interior Lighting	130128.51
Interior Equipment	130128.51
Fans	2612.87
Total Electricity Consumption	359939.78

Table 8: Annual End-Use Electricity Consumption for Navrangpura Retrofit Model

The simulation indicates that cooling, interior lighting and interior equipment constitute the major energy-consuming end uses within the district, while fan energy represents only a small fraction of the total electricity consumption.

13.2.9 Thaltej District

The same retrofit measures were applied to the Thaltej district model. The resulting annual energy performance obtained from the EnergyPlus simulation is summarized in Table 13.2.9.

Parameter	Value
Total Building Area (m ²)	1191137.13
Conditioned Area (m ²)	595568.57
Total Site Energy (GJ)	537968.87
Site Energy Intensity (MJ/m ²)	451.64
Total Source Energy (GJ)	1703747.42
Source Energy Intensity (MJ/m ²)	1430.35

Table 9: Annual Energy Performance of Thaltej Retrofit Model

End Use	Energy (GJ)
Cooling	158091.95
Interior Lighting	187818.50
Interior Equipment	187818.50
Fans	4239.92
Total Electricity Consumption	537968.87

Table 10: Annual End-Use Electricity Consumption for Thaltej Retrofit Model

Similar to the Navrangpura district, the retrofit simulation for Thaltej shows that cooling, lighting and interior equipment dominate the annual electricity demand. The larger building stock and conditioned floor area of Thaltej result in higher total annual energy consumption compared to Navrangpura.

13.3 Comparison of Baseline and Retrofit Simulation Results

To evaluate the effectiveness of the proposed retrofit strategies, the simulation results obtained from the retrofit models were compared with the baseline energy performance for both study areas. The comparison focuses on the annual site energy, source energy, energy intensity, and major end-use electricity consumption. The percentage reduction for each parameter was calculated using

$$Reduction(\%) = \frac{Baseline - Retrofit}{Baseline} \times 100 \quad (3)$$

The comparison provides a quantitative assessment of the energy savings achieved through improvements to the building envelope, glazing system, cooling system efficiency, and lighting power density.

Navrangpura District

Parameter	Baseline	Retrofit	Reduction (%)
Total Site Energy (GJ)	429048.28	359939.78	16.11
Total Source Energy (GJ)	1358795.89	1139929.29	16.11
Site Energy Intensity (MJ/m ²)	519.89	436.15	16.11
Source Energy Intensity (MJ/m ²)	1646.49	1381.28	16.11
Cooling Energy (GJ)	139588.60	97069.89	30.46
Interior Lighting (GJ)	156154.21	130128.51	16.67
Interior Equipment (GJ)	130128.51	130128.51	0.00
Fan Energy (GJ)	3176.96	2612.87	17.75

Table 11: Comparison of Baseline and Retrofit Energy Performance for Navrangpura

The retrofit measures reduced the total annual site energy consumption of the Navrangpura district by approximately 16.1%, while cooling energy consumption decreased by about 30.5%. The substantial reduction in cooling demand demonstrates the effectiveness of the improved building envelope, high-performance glazing, and increased cooling system efficiency. Interior lighting energy also decreased owing to the reduction in lighting power density, whereas interior equipment energy remained unchanged because equipment schedules and loads were not modified in the retrofit model.

Thaltej District

Parameter	Baseline	Retrofit	Reduction (%)
Total Site Energy (GJ)	651649.56	537968.87	17.44
Total Source Energy (GJ)	2063774.14	1703747.42	17.44
Site Energy Intensity (MJ/m ²)	547.08	451.64	17.44
Source Energy Intensity (MJ/m ²)	1732.61	1430.35	17.44
Cooling Energy (GJ)	233170.63	158091.95	32.20
Interior Lighting (GJ)	225382.20	187818.50	16.67
Interior Equipment (GJ)	187818.50	187818.50	0.00
Fan Energy (GJ)	5278.22	4239.92	19.67

Table 12: Comparison of Baseline and Retrofit Energy Performance for Thaltej

The retrofit strategy achieved an overall reduction of approximately 17.4% in annual site energy consumption for the Thaltej district. Cooling energy exhibited the largest improvement, with a reduction of approximately 32.2%, confirming that enhancements to the thermal envelope and cooling system are particularly effective for Ahmedabad's cooling-dominated climate. Similar to the Navrangpura model, lighting energy consumption decreased due to the reduced lighting power density, while interior equipment loads remained unchanged.

The comparison between the baseline and retrofit simulations demonstrates that the proposed retrofit measures significantly improve the annual energy performance of residential buildings in both study areas. Although Thaltej has a larger building stock and consequently higher total energy consumption than Navrangpura, both districts exhibit similar trends in energy savings after retrofit implementation.

Among all end uses, cooling energy experienced the greatest reduction, ranging from approximately 30–32%, highlighting the importance of improving the thermal performance of the building envelope in hot climatic regions. The reductions in wall, roof and window U-values decreased conductive heat gains, while the improved cooling system efficiency further lowered electricity consumption. In addition, reducing the lighting power density contributed to lower lighting energy use and indirectly reduced cooling loads by decreasing internal heat gains.

Overall, the retrofit strategies reduced annual site energy consumption by approximately 16.1% for Navrangpura and 17.4% for Thaltej. These results demonstrate that integrating external PUF insulation, high-performance glazing, improved HVAC efficiency and efficient lighting provides a practical and effective approach for enhancing the energy performance of existing residential buildings in Ahmedabad.

14 Challenges Faced

Throughout the internship, several technical challenges were encountered during the preparation of CityBES-compatible datasets for UBEM. These challenges were addressed through literature review, software exploration and regular discussions with the project mentor.

1. Availability of Building Height Information

One of the major challenges was the limited availability of building height information in publicly available OpenStreetMap datasets. Although building footprints were available for several regions of Ahmedabad, essential attributes such as `building:levels` were missing for many areas including Bopal and Naroda. Since building height is a mandatory input for CityBES, these datasets could not be directly used for Urban Building Energy Modeling.

Different approaches for obtaining building heights were explored, including the Ahmedabad 3D City Model. However, manual extraction of heights for a large number of buildings was found to be impractical. Consequently, the study areas were limited to Navrangpura and Thaltej, where sufficient building-level information was available, and building heights were estimated using the available number of storeys.

$$Height = NumberofStoreys \times 3m$$

This approach provided consistent height estimates suitable for Urban Building Energy Modeling.

2. Selection of an Appropriate Weather File

Another challenge was selecting a suitable EnergyPlus Weather (EPW) file for Ahmedabad. The official EnergyPlus weather database primarily contained older weather datasets, which were not considered ideal for the project.

Alternative repositories, including Climate.OneBuilding, were explored to obtain updated Typical Meteorological Year (TMYx) weather files. Multiple EPW files were compared based on their temporal coverage and compatibility with CityBES before selecting the most appropriate weather file for the simulations.

3. Preparation of CityBES-Compatible GeoJSON Datasets

Preparing datasets compatible with CityBES was one of the major challenges encountered during the internship. Although QGIS allows direct export of GeoJSON files, the exported datasets required modifications to satisfy the specific input structure expected

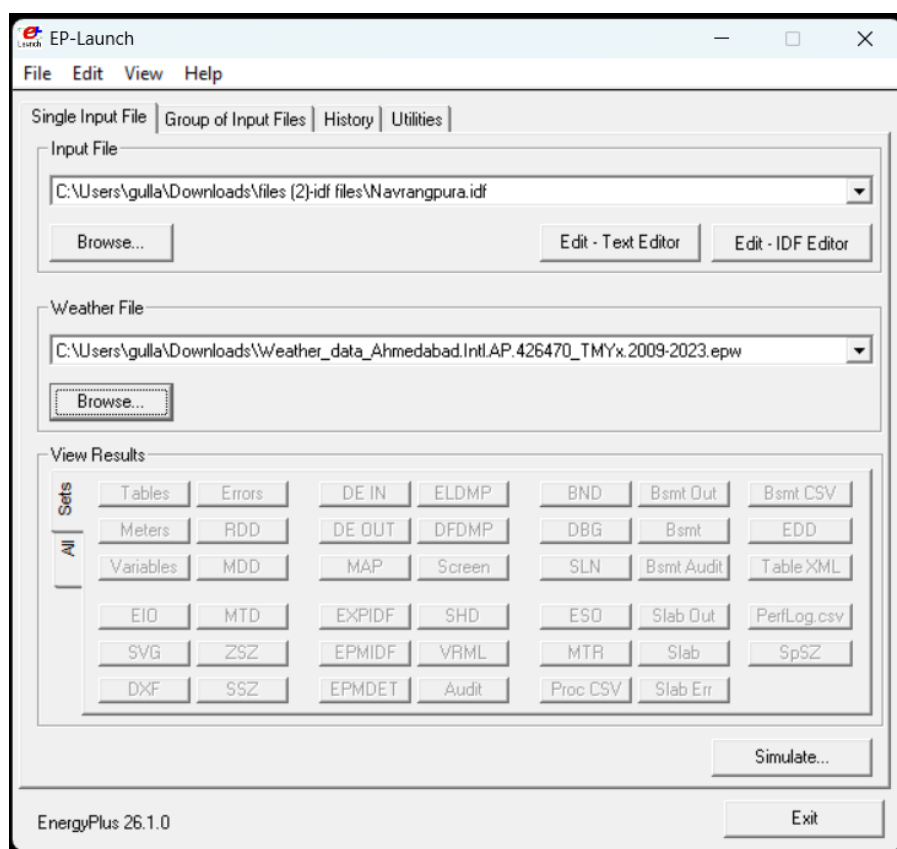


Figure 15: Energy Plus (UDEM Engine)

by CityBES. This involved understanding the required GeoJSON syntax, mandatory building properties, coordinate reference system requirements, and the structure of the `additional_json` object.

Initially, it was assumed that the default CityBES building prototypes would be sufficient for the simulations. However, detailed study of the CityBES documentation and discussions with the project mentor revealed that these default assumptions are primarily based on ASHRAE standards representing North American buildings. Consequently, variations in operational parameters, particularly heating and cooling thermostat set-points and schedules, were identified for Ahmedabad's hot climatic conditions.

To address this challenge, the required operational parameters were first identified by comparing the default CityBES prototype assumptions with Ahmedabad residential building characteristics. Subsequently, Python scripts were developed and executed in Visual Studio Code to automatically modify the exported GeoJSON files by appending an `additional_json` object containing the selected Ahmedabad-specific parameters for every building. This automated approach ensured consistency across the datasets while eliminating the need for manual editing of hundreds of building records.

4. High Computational Time for Energy Simulations

One of the major challenges encountered during the project was the long computational time required for EnergyPlus simulations. The baseline simulations took approximately 20 hours to complete for each study area (Navrangpura and Thaltej) due to the large number of buildings included in the district-scale models. Even after implementing the retrofit measures, the simulation time remained significant, requiring approximately 6 hours for Navrangpura and 8 hours for Thaltej. Since each modification to the building model required a complete re-simulation, validating retrofit strategies and correcting input errors became a time-intensive process, increasing the overall project duration.

15 Learning Outcomes

The internship provided valuable exposure to Geographic Information Systems (GIS), Urban Building Energy Modeling (UBEM), building energy simulation workflows, and technical documentation. Both technical and professional skills were developed throughout the project.

Technical Skills

- Developed a comprehensive understanding of Urban Building Energy Modeling (UBEM) and its applications in city-scale energy analysis.
- Learned the complete workflow of preparing CityBES-compatible datasets from raw GIS data.
- Gained practical experience in QGIS for spatial data processing, attribute management, field calculations, spatial visualization, and GeoJSON generation.
- Learned to extract building datasets from OpenStreetMap using the QuickOSM plugin and explored alternative geospatial data sources including VEDAS, Bhuvan, Google Earth Pro, Ahmedabad 3D City Model, and Copernicus Browser.
- Performed GIS data preprocessing, including attribute cleaning, geometry validation, building height estimation, and building classification based on footprint area.
- Developed an understanding of the GeoJSON data structure and learned how CityBES maps building attributes to EnergyPlus simulation inputs.
- Acquired proficiency in Python for automating GeoJSON modifications and appending Ahmedabad-specific operational parameters through the `additional_json` field.
- Learned the significance of EnergyPlus Weather (EPW) files and the process of selecting suitable weather datasets for building energy simulations.
- Gained knowledge of building energy concepts including HVAC systems, thermostat setpoints, operational schedules, window properties, thermal comfort, insulation, and building envelope characteristics.
- Learned the workflow of EnergyPlus simulations using EP-Launch, including the role of IDF (Input Data File) and EPW (EnergyPlus Weather File) in executing building energy simulations.
- Understood the relationship between CityBES and EnergyPlus, where CityBES generates simulation-ready EnergyPlus models for city-scale energy analysis.

- Developed the ability to interpret technical documentation and implement software-specific input requirements.
- Learned to perform baseline and retrofit energy simulations using EnergyPlus and interpret the resulting annual building energy performance metrics.
- Gained practical experience in evaluating the impact of Energy Conservation Measures (ECMs) by comparing baseline and retrofit simulation results using key performance indicators such as site energy, source energy, energy intensity, and end-use electricity consumption.
- Developed an understanding of thermal transmittance (U-value) calculations based on ISO 6946 and their role in improving building envelope performance during retrofit design.
- Learned how modifications to building envelope properties, glazing systems, HVAC efficiency, and lighting power density influence annual cooling loads and overall building energy consumption.
- Acquired experience in validating simulation outputs and interpreting EnergyPlus summary reports to identify dominant building end uses and quantify potential energy savings.
- Learned to interpret EnergyPlus simulation outputs and evaluate building energy performance using key performance indicators such as Site Energy, Source Energy, Energy Use Intensity (EUI), end-use energy consumption, and operational carbon emissions.

Professional Skills

- Strengthened analytical thinking and engineering problem-solving skills through systematic evaluation of multiple data sources and modeling approaches.
- Improved technical documentation and report writing skills.
- Enhanced programming skills by developing Python scripts for workflow automation.
- Improved communication and collaboration through regular discussions with mentors and team members.
- Developed project planning and time management skills by completing sequential phases of a real-world engineering project.
- Learned to independently explore new software tools, technical documentation, and research resources.

- Understood the importance of data quality, verification, and standardized workflows in engineering applications.
- Improved debugging and troubleshooting skills while resolving issues related to GIS data processing, GeoJSON formatting, and software compatibility.
- Developed patience and computational workflow management skills while executing long-duration EnergyPlus simulations and validating multiple retrofit scenarios.

16 Future Scope

The present work focused on developing a workflow for city-scale energy modeling and evaluating retrofit strategies for selected residential areas in Ahmedabad. Several opportunities exist for extending and improving the scope of this work in the future.

- **City-wide Analysis:** Extend the developed workflow to cover the entire Ahmedabad city or other Indian cities, enabling large-scale assessment of building energy performance and retrofit potential.
- **Inclusion of Additional Building Types:** Expand the study to include commercial, institutional, educational, healthcare, and mixed-use buildings to obtain a more comprehensive understanding of urban energy consumption.
- **Gap Analysis Between Simulated and Actual Energy Consumption:** Compare simulation results with actual building electricity consumption data to validate the models, identify performance gaps, and improve simulation accuracy.
- **Evaluation of Additional Energy Conservation Measures (ECMs):** Investigate the impact of advanced retrofit measures such as cool roofs, green roofs, photovoltaic (PV) systems, natural ventilation strategies, energy storage systems, and smart building controls.
- **Artificial Intelligence and Machine Learning Integration:** Apply AI and machine learning techniques to predict building energy consumption, optimize retrofit selection, and identify the most cost-effective energy conservation strategies.
- **Development of an Interactive Decision Support Platform:** Create a web-based dashboard integrating GIS visualization and simulation results to assist planners, policymakers, and engineers in evaluating district-scale retrofit scenarios.
- **Economic and Environmental Assessment:** Incorporate life-cycle cost analysis, payback period estimation, and carbon emission reduction analysis to evaluate the economic feasibility and environmental benefits of retrofit strategies.

The workflow developed during this internship provides a strong foundation for future city-scale building energy studies and can support sustainable urban planning and energy-efficient retrofit decision-making.

17 Conclusion

This internship successfully achieved the objective of developing a complete workflow for city-scale building energy modeling and retrofit analysis using GIS, CityBES, and EnergyPlus. Beginning with the collection and preprocessing of geospatial building data, the project involved preparing CityBES-compatible datasets, generating EnergyPlus simulation models, performing baseline energy simulations, implementing multiple Energy Conservation Measures (ECMs), and evaluating the effectiveness of retrofit strategies through comparative energy analysis.

The baseline simulations identified cooling, interior lighting, and interior equipment as the dominant contributors to annual building energy consumption in both Navrangpura and Thaltej. Based on these findings, appropriate retrofit measures, including external Polyurethane Foam (PUF) insulation, Vacuum Insulated Glazing (VIG), improved cooling system efficiency, and reduced lighting power density, were implemented. The retrofit simulations demonstrated significant improvements in building energy performance. Annual site and source energy consumption were reduced by approximately **16.1%** in Navrangpura and **17.4%** in Thaltej, while cooling energy demand decreased by approximately **30.5%** and **32.2%**, respectively. These improvements also resulted in lower Energy Use Intensity (EUI) and operational carbon emissions, demonstrating the effectiveness of the proposed retrofit measures for residential buildings under Ahmedabad's hot climatic conditions.

Throughout the internship, valuable experience was gained in GIS, UDEM, EnergyPlus simulations, Python automation, and building energy analysis. The project also strengthened technical documentation, analytical thinking, engineering problem-solving, and project management skills while providing practical exposure to real-world building energy modeling applications.

Overall, the internship provided a comprehensive understanding of the complete workflow involved in city-scale building energy analysis, from GIS data preparation and simulation model development to retrofit implementation and performance evaluation. The developed workflow establishes a scalable framework for evaluating building energy performance and demonstrates how data-driven retrofit strategies can significantly reduce energy consumption and operational carbon emissions while supporting sustainable urban development. The knowledge and practical experience gained during this internship provide a strong foundation for future work in the fields of Urban Building Energy Modeling, building energy simulation, sustainable buildings, and climate-responsive urban planning.

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2. F. P. Incropera, D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, John Wiley & Sons
3. ASHRAE, *ASHRAE Handbook – Fundamentals*, American Society of Heating, Refrigerating and Air-Conditioning Engineers

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1. City Building Energy Saver (CityBES) User Guide
2. CityBES Data Preparation Guide
3. EnergyPlus Documentation
4. EP-Launch User Manual
5. QGIS User Guide Version 3.40

6. OpenStreetMap Documentation
7. Python 3.12 Documentation

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2. Lawrence Berkeley National Laboratory
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<https://climate.onebuilding.org>
5. OpenStreetMap
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6. QGIS Documentation
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19 Appendices

Navrangpura

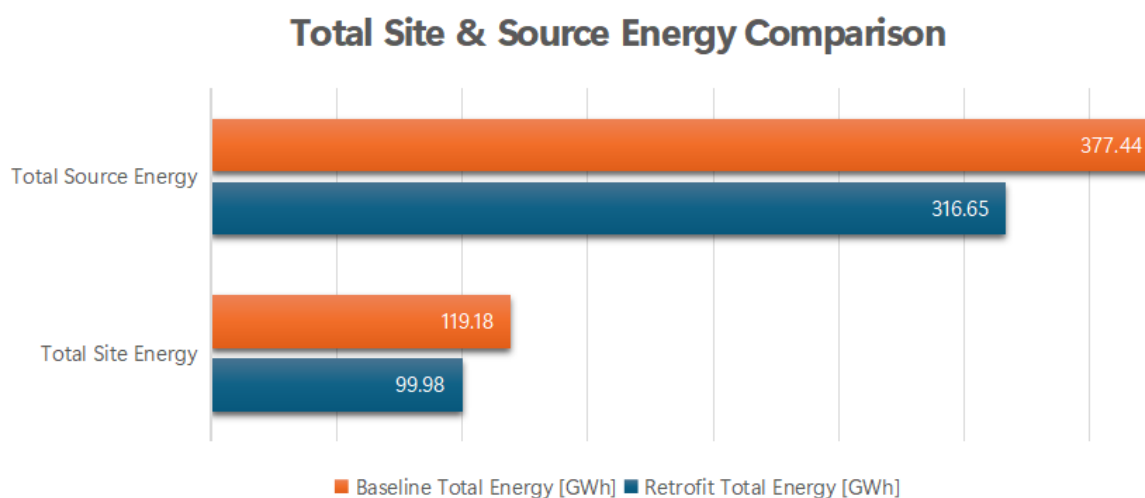


Figure 16: Comparison of Annual Site and Source Energy for Baseline and Retrofit Models (Navrangpura District)

Figure 16 compares the total annual site and source energy consumption of the baseline and retrofit models for the Navrangpura district. The retrofit measures reduced the total site energy from 119.18 GWh to 99.98 GWh and the total source energy from 377.44 GWh to 316.65 GWh, corresponding to an overall reduction of approximately 16.11%. These reductions demonstrate the effectiveness of the proposed retrofit measures in lowering the overall annual energy demand.

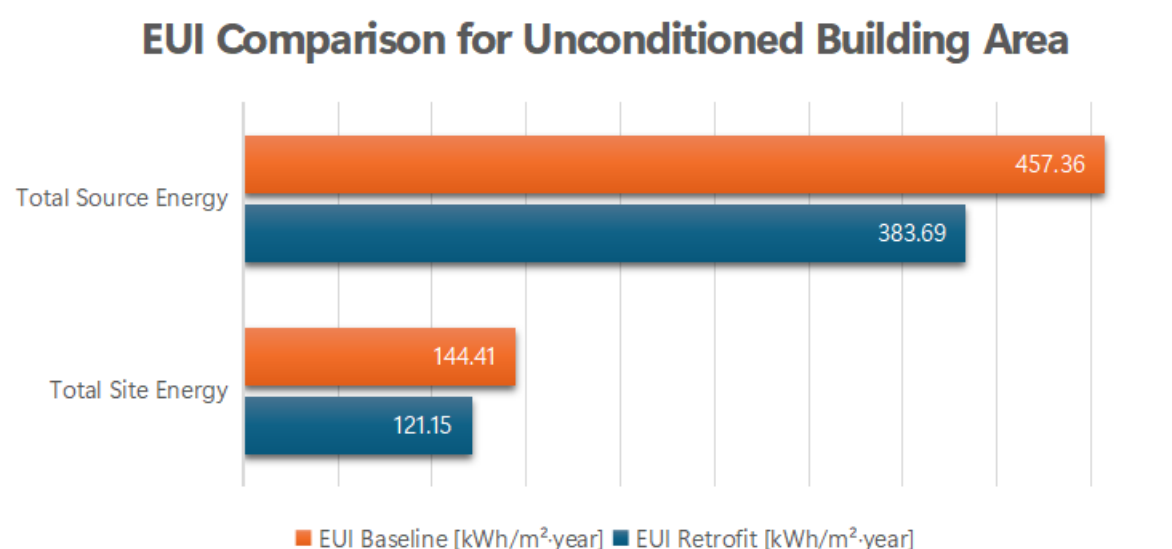


Figure 17: Comparison of Energy Use Intensity (Total Building Area) for Baseline and Retrofit Models (Navrangpura District)

Figure 17 illustrates the Energy Use Intensity (EUI) normalized by the total building area. The retrofit reduced the site EUI from 144.41 to 121.15 kWh/m²-year and the source EUI from 457.36 to 383.69 kWh/m²-year, demonstrating improved overall energy performance at the district scale.

EUI Comparison for Conditioned Building Area

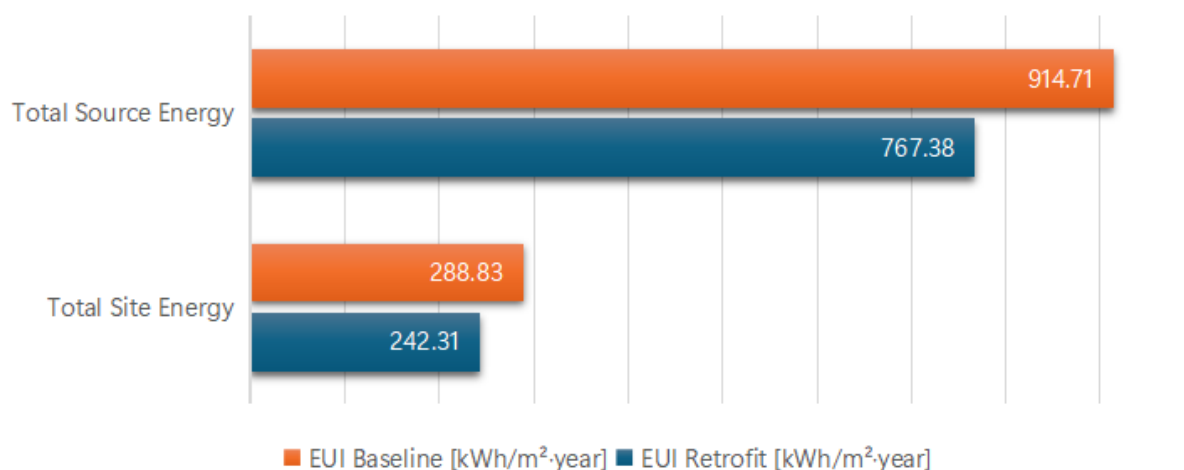


Figure 18: Comparison of Energy Use Intensity (Conditioned Building Area) for Baseline and Retrofit Models (Navrangpura District)

Figure 18 compares the annual Energy Use Intensity (EUI) for the conditioned building area. The retrofit reduced the site EUI from 288.83 to 242.31 kWh/m²-year and the source EUI from 914.71 to 767.38 kWh/m²-year. These reductions indicate an overall improvement in the energy efficiency of conditioned residential buildings after implementing the retrofit measures.

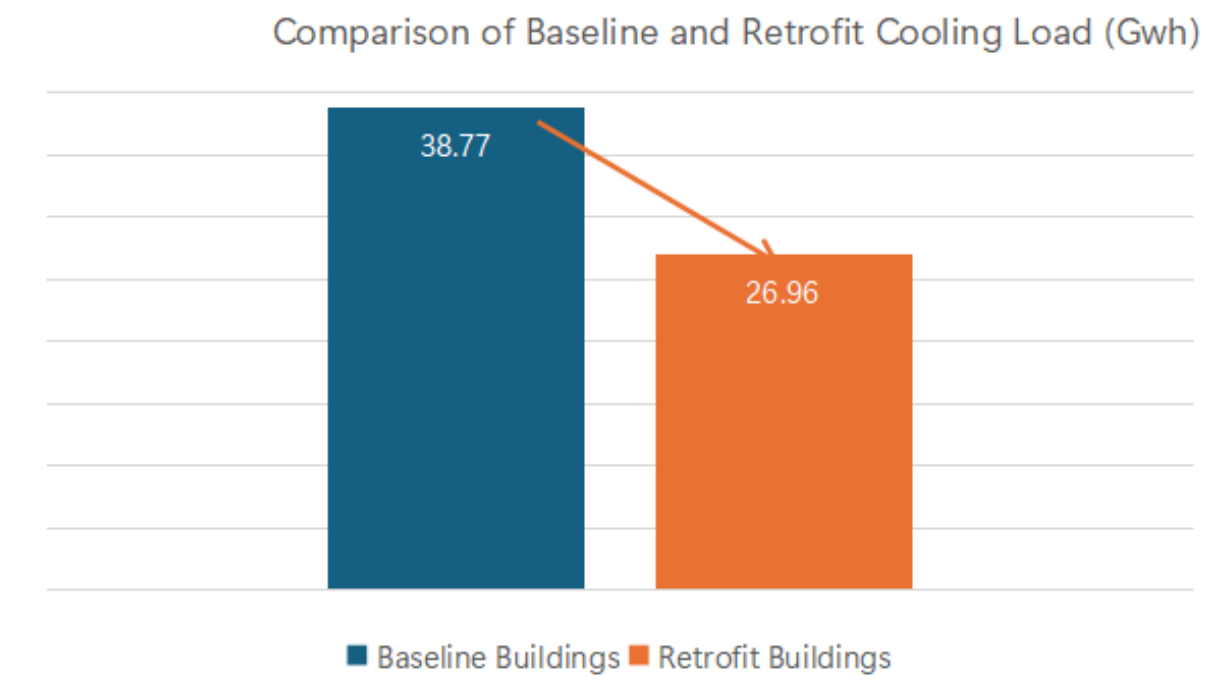


Figure 19: Comparison of Annual Cooling Energy Consumption for Baseline and Retrofit Models (Navrangpura District)

Figure 19 compares the annual cooling energy demand before and after retrofit implementation. Cooling energy decreased from 38.77 GWh for the baseline buildings to 26.96 GWh after retrofit, corresponding to a reduction of approximately 30.46%. This significant reduction confirms that improvements to the building envelope, glazing system, and cooling system efficiency effectively reduced the cooling demand under Ahmedabad's hot climatic conditions.

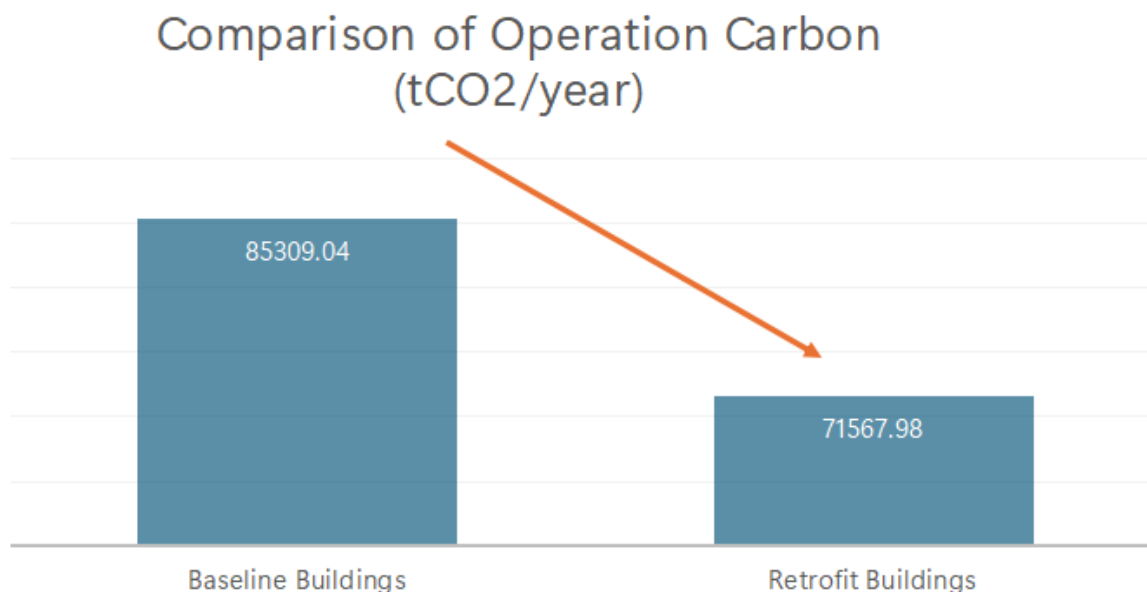


Figure 20: Comparison of Annual Operational Carbon Emissions for Baseline and Retrofit Models (Navrangpura District)

Figure 20 presents the annual operational carbon emissions estimated for the baseline and retrofit models. The retrofit reduced operational carbon emissions from 85,309.04 tCO₂/year to 71,567.98 tCO₂/year, representing an overall reduction of approximately 16.11%. The decrease in carbon emissions reflects the reduction in annual electricity consumption achieved through the implemented retrofit measures and highlights the environmental benefits of improving building energy efficiency.

Thaltej

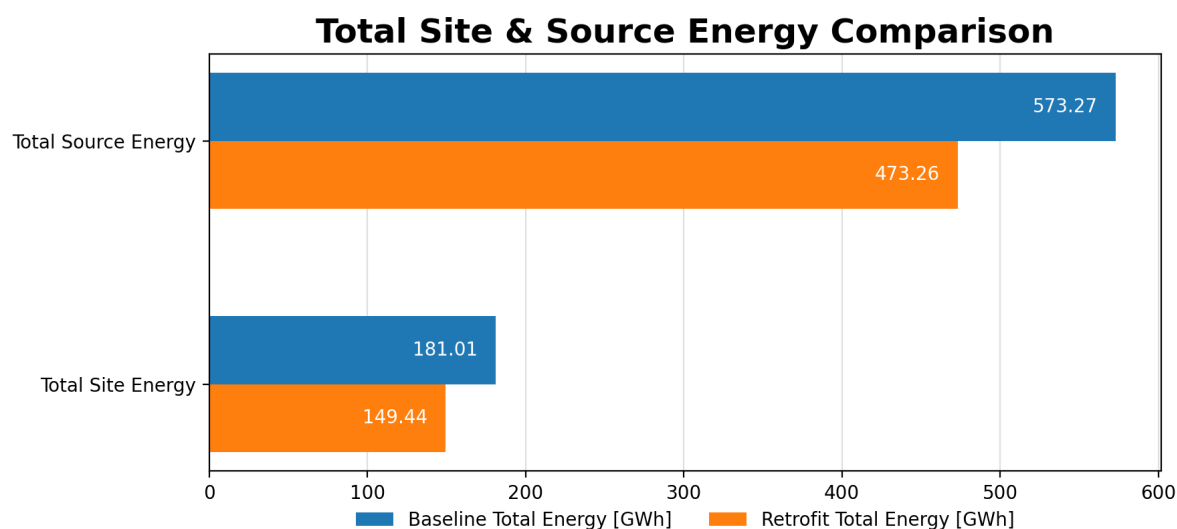


Figure 21: Comparison of Annual Site and Source Energy for Baseline and Retrofit Models (Thaltej District)

Figure 21 compares the total annual site and source energy consumption of the baseline and retrofit models for the Thaltej district. The retrofit measures reduced the total site energy from 181.01 GWh to 149.44 GWh and the total source energy from 573.27 GWh to 473.26 GWh, corresponding to an overall reduction of approximately 17.44%. These reductions demonstrate the effectiveness of the proposed retrofit measures in lowering the overall annual energy demand.

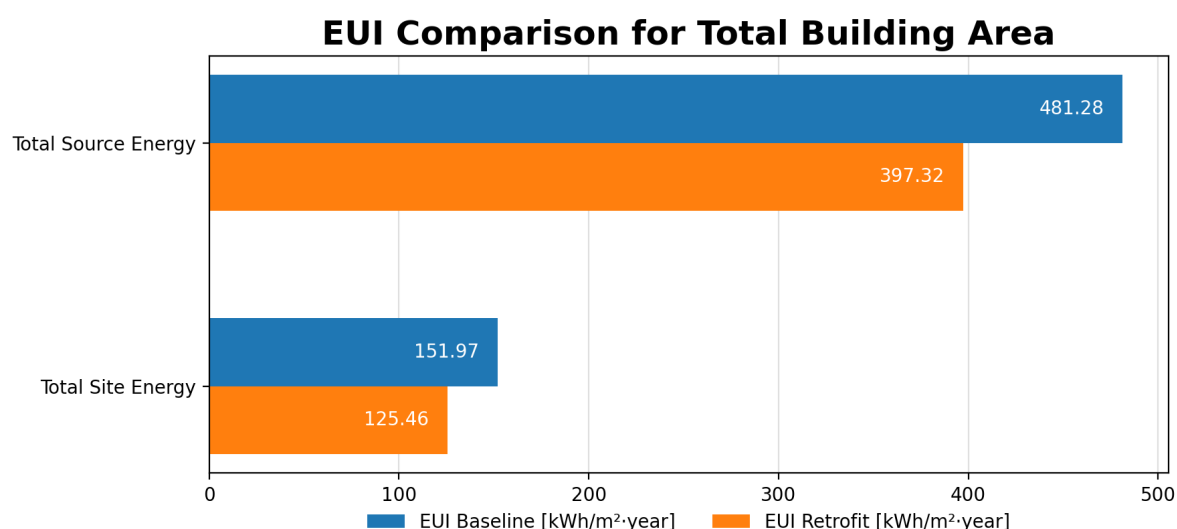


Figure 22: Comparison of Energy Use Intensity (Total Building Area) for Baseline and Retrofit Models (Thaltej District)

Figure 22 illustrates the Energy Use Intensity (EUI) normalized by the total building

area. The retrofit reduced the site EUI from 151.97 to 125.46 kWh/m²·year and the source EUI from 481.28 to 397.32 kWh/m²·year, demonstrating improved overall energy performance at the district scale.

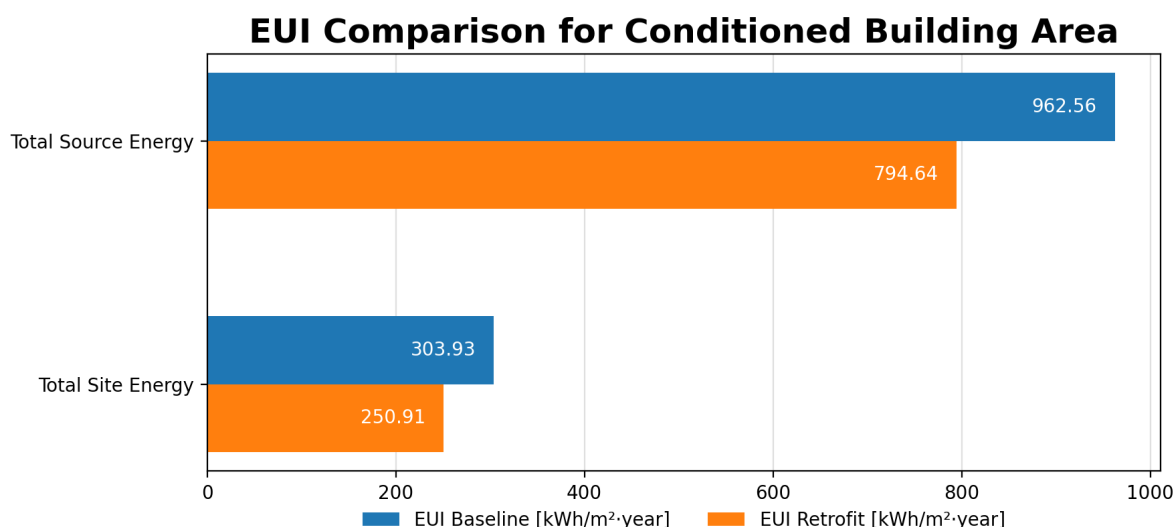


Figure 23: Comparison of Energy Use Intensity (Conditioned Building Area) for Baseline and Retrofit Models (Thaltej District)

Figure 23 compares the annual Energy Use Intensity (EUI) for the conditioned building area. The retrofit reduced the site EUI from 303.93 to 250.93 kWh/m²·year and the source EUI from 962.55 to 794.63 kWh/m²·year. These reductions indicate an overall improvement in the energy efficiency of conditioned residential buildings after implementing the retrofit measures.

Comparison of Baseline and Retrofit Cooling Load (GW)

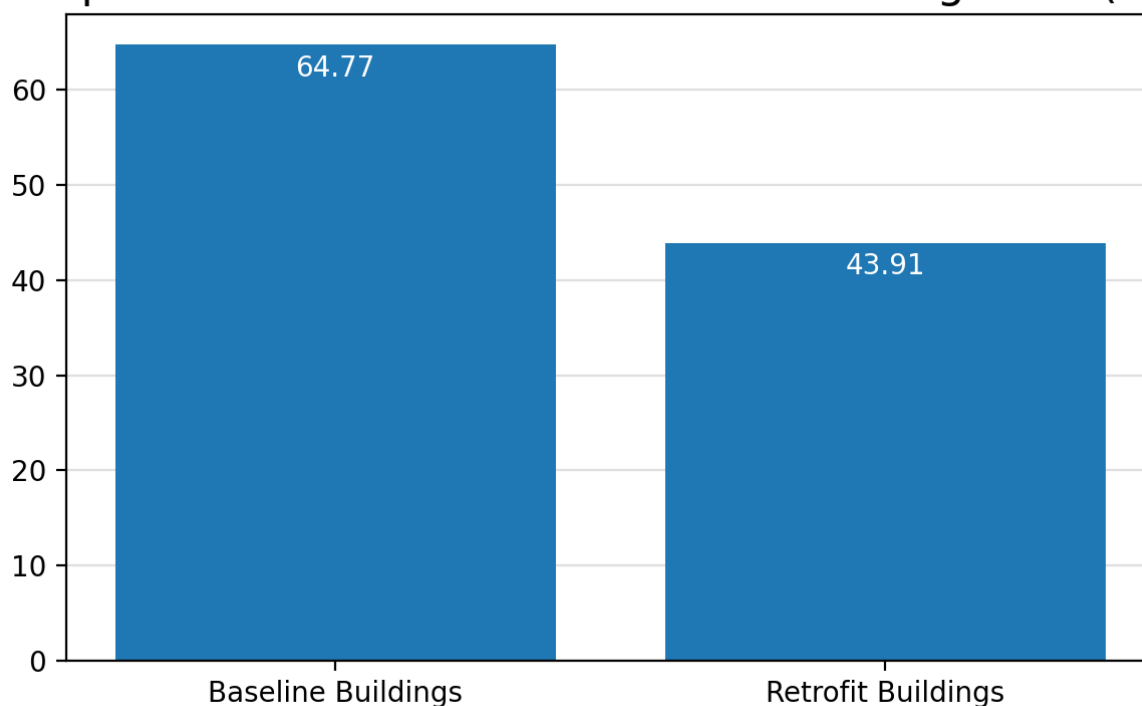


Figure 24: Comparison of Annual Cooling Energy Consumption for Baseline and Retrofit Models (Thaltej District)

Figure 24 compares the annual cooling energy demand before and after retrofit implementation. Cooling energy decreased from 64.77 GWh for the baseline buildings to 43.91 GWh after retrofit, corresponding to a reduction of approximately 32.20%. This significant reduction confirms that improvements to the building envelope, glazing system, and cooling system efficiency effectively reduced the cooling demand under Ahmedabad's hot climatic conditions.

Weekly Progress Report

Week	Activities Completed	Skills Learned
1	Project orientation and literature review on Urban Building Energy Modeling (UBEM), CityBES, EnergyPlus, building envelope, insulation materials, thermal comfort, energy retrofitting, and energy benchmarking.	UBEM concepts, building energy fundamentals, CityBES workflow, literature review.
2	Explored GIS data sources including OpenStreetMap, VEDAS, Bhuvan, Google Earth Pro, and Copernicus Browser. Extracted residential building footprints for different regions of Ahmedabad using QGIS and QuickOSM.	QGIS basics, GIS data extraction, OpenStreetMap, geospatial data exploration.
3	Processed building datasets by cleaning attribute tables, estimating building heights from building levels, calculating footprint and floor areas, and selecting Navrangpura and Thaltej as the study areas.	GIS data preprocessing, attribute management, field calculations, height estimation.
4	Classified buildings into Small, Medium, and Large categories for visualization. Explored QGIS 3D visualization and selected the appropriate Ahmedabad EPW weather file after comparing multiple weather data sources.	Building classification, data visualization, EnergyPlus weather (EPW) files, climate data selection.
5	Prepared CityBES-compatible GeoJSON datasets by studying the required GeoJSON structure, mandatory attributes, coordinate reference systems, and CityBES input specifications.	GeoJSON preparation, JSON syntax, CityBES data requirements.
6	Compared CityBES default prototype assumptions with Ahmedabad residential building characteristics and developed Python scripts to automatically append <code>additional_json</code> parameters to every building.	Python programming, JSON manipulation, workflow automation, CityBES parameter customization.
7	Reviewed and finalized the datasets by incorporating HVAC configuration, operational schedules, thermostat setpoints, and weather inputs. Verified compatibility with CityBES before simulation.	Dataset verification, debugging, EnergyPlus input parameters, quality assurance.
8	Uploaded the prepared datasets to the CityBES platform, performed baseline building energy simulations, analyzed the generated results, and completed the project documentation and internship report.	CityBES simulation workflow, result interpretation, technical documentation, engineering report writing.

Table 13: Weekly Progress Report